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Enhancing cybersecurity via attribute reduction with deep learning model for false data injection attack recognition

Faheed A.F. Alrslani[1], Manal Abdullah Alohali[2]✉, Mohammed Aljebreen[3], Hamed Alqahtani[4], Asma Alshuhail[5], Menwa Alshammeri[6] & Wafa Sulaiman Almukadi[7]

Cyberattacks have given rise to several phenomena and have raised concerns among users and power system operators. When they are built to bypass state estimation bad data recognition methods executed in the conventional grid system control room, False Data Injection Attacks (FDIA) pose a significant security threat to the operation of power systems. Therefore, real-time monitoring becomes inevitable with the quick implementation of renewables within the grid operator. The state estimation algorithm plays a major role in defining the grid's operating scenarios. FDIA creates a significant risk to these estimation strategies by adding malicious information to the measurement obtained. Real-time recognition of these attack classes improves grid resiliency and ensures a secure grid operation. This study introduces a novel Attribute Reduction with a Deep Learning-based False Data Injection Attack Recognition (ARDL-FDIAR) technique. The primary goal of the ARDL-FDIAR technique is to improve security via the FDIA detection process. The ARDL-FDIAR technique uses Z-score normalization to scale the input data. The attribute reduction process gets invoked using the modified Lemrus optimization algorithm (MLOA) to choose optimal feature sets. Moreover, the FDIA detection process is performed by modelling an improved deep belief network (IDBN) model. Furthermore, the performance of the IDBN model is improved by the Cetacean Optimization Algorithm (COA)-based hyperparameter tuning process. A series of experiments were performed to ensure the enhancement of the ARDL-FDIAR technique. The results indicated the enhanced security performance of the ARDL-FDIAR technique compared to other DL approaches.

Keywords False data injection attack, Cyberattack, Deep learning, Cetacean optimization Algorithm, Deep Belief Network

A model of smart, sustainable cities has been developed currently, targeting to improve the quality of life, enhance the city process, and integrate renewable energy resources to be further globally supportable[1]. The widespread use of information and communication technologies (ICT) was a main promoter in attaining the objective of smart ecological cities. For instance, new power methods quickly alter near the smart grid, conveying an energy delivery system by a dual-way stream of power and data. By combining numerous innovative communication technologies, the power method is affecting near the way of the smart grid[2]. Conversely, the power grid is meeting numerous safety challenges owing to the addition of cyberspace to the physical space. Furthermore, substantial real power method data has taken progressive ability and the task of defending smart grid methods[3]. Physical and cyber securities are dual significant power features of system safety. Physical security can uphold

[1]Department of Information Technology, Faculty of Computing and Information Technology, Northern Border University, Rafha, Saudi Arabia. [2]Department of Information Systems, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia. [3]Department of Computer Science, Community College, King Saud University, P.O. Box 28095, Riyadh 11437, Saudi Arabia. [4]Department of Information Systems, College of Computer Science, Center of Artificial Intelligence, Unit of Cybersecurity, King Khalid University, Abha, Saudi Arabia. [5]Department of Information Systems, College of Computer Sciences & Information Technology, King Faisal University, Hofuf, Saudi Arabia. [6]Department of Computer Science, College of Computer and Information Sciences, Jouf University, Sakaka, Saudi Arabia. [7]Department of Software Engineering, College of Engineering and Computer Science, University of Jeddah, Jeddah, Saudi Arabia. ✉email: maalohaly@pnu.edu.sa

constant supply during equipment failures[4]. Cybersecurity denotes the safety of the SCADA model, which upholds the power method process. In the present scenario, cyber-attacks have slowly exposed current power methods owing to the global usage of communication technologies[5]. Also, due to the adjacent linking between SCADA and physical methods, physical safety cooperated with cybersecurity exposures.

Any disturbance or potential attack on the smart grid's functioning is dangerous because electricity plays a crucial role in everyday life[6]. Moreover, many social organizations (e.g., hospitals) and trade devices (e.g., manufacturing plants) primarily run on electric control. The False Data Injection Attack (FDIA) cooperates with the reliability of data (meters, sensors, etc.), where both the attacker and environment work as an actual value with a false value[7]. Based on the kind of effective FDIA, it will destroy the physical structure or outcome in the defeat of the economy and, vitally, in numerous cases, it is dangerous. Therefore, the measure of damage perceiving and averting FDIA in the power model is of great significance[8]. Deep learning (DL) techniques are currently recommended to grab the greater order numerical structure of the intricate data over arranging the detector feature in layers. One of the vital DL models commonly utilized is the deep belief network (DBN), which is generated with a set of restricted Boltzmann machines (RBM)[9]. Selecting suitable variables to execute the neural networks (NNs) effectively. The control of MGs is generally implemented utilizing power, and present measurements, and potential attackers believe them as straight objectives[10]. Over-interminating fault data to the voltage and present measurements will present the attacked variable to monitor the application, and it can also disorder MG application control.

This study introduces a novel Attribute Reduction with a Deep Learning-based False Data Injection Attack Recognition (ARDL-FDIAR) technique. The primary goal of the ARDL-FDIAR technique is to improve security via the FDIA detection process. The ARDL-FDIAR technique uses Z-score normalization to scale the input data. The attribute reduction process gets invoked using the modified Lemrus optimization algorithm (MLOA) to choose optimal feature sets. Moreover, the FDIA detection process is performed by modelling an improved deep belief network (IDBN) model. Furthermore, the performance of the IDBN model is improved by the Cetacean Optimization Algorithm (COA)-based hyperparameter tuning process. A series of experiments were performed to ensure the enhancement of the ARDL-FDIAR technique. The key contribution of the ARDL-FDIAR technique is listed below.

- The Z-score normalization is utilized to standardize the input data, which substantially improves the stability and performance of the technique by confirming that the overall features are on a similar scale. This preprocessing step mitigates the impact of outliers and enhances convergence during training. The model becomes more robust and efficient in learning patterns from the input by converting the data to have a mean of zero and a standard deviation of one.
- The MLOA approach is employed for optimal feature selection, effectually mitigating dimensionality while improving the relevance of input features. This methodology minimizes redundancy and noise, allowing the model to concentrate on the most impactful attributes. By streamlining the feature set, the MLOA substantially enhances the accuracy and efficiency of the model in detecting false data injection attacks (FDIA).
- A robust FDIA detection process was developed using an IDBN model, which employs advanced learning capabilities for more accurate detection. This methodology efficiently captures intrinsic patterns within the data, improving its ability to differentiate between legitimate and malevolent inputs. By incorporating DL methods, the IDBN crucially enhances the reliability and precision of FDIA detection in dynamic environments.
- The IDBN model's performance was substantially improved through hyperparameter tuning by employing the COA method, which systematically fine-tunes model parameters for optimal performance. This methodology confirms that the model operates at its highest potential, enhancing detection accuracy in detecting FDIA. By employing COA, the model benefits from a more tailored approach to parameter optimization, resulting in improved overall robustness and effectiveness.
- Incorporating the MLOA for feature selection and the COA for hyperparameter tuning within the IDBN framework presents a novel method for enhancing FDIA detection. This integration of advanced optimization models streamlines the input features and confirms that the model parameters are finely tuned for optimum performance. By concurrently addressing feature relevance and model optimization, this methodology significantly improves detection accuracy and robustness in complex datasets. This dual focus on optimization depicts a crucial enhancement in the field, setting a new benchmark for FDIA detection systems.

The remaining sections of the article are arranged as follows: Sect. "Literature review" offers the literature review, and Sect. "Proposed methodology" represents the proposed method. Then, Sect. "Experimental validation" elaborates on the results evaluation, and Sect. "Conclusion" completes the work.

Literature review

Tian et al.[11] discover the multi-label adversarial sample assaults beside multi-label FDIA location detectors and suggest a common structure of multi-label adversarial attacks, i.e. multi-label adversarial FDIA (LESSON). The developed LESSON attack structures contain 3 three main designs, i.e. Tailored Loss Function Design, Perturbing State Variables, and Change of Variable, that aid in discovering appropriate multi-label argumentative perturbations in the physical restraints. In[12], a spatio-temporal graph DL (STGDL)-based scheme is projected. Initially, the graph and time-based gated convolutional were coordinated to remove spatio-temporal features. Next, a quantile regression training (QRT) tactic was assumed to provide operative limits variable in state estimation (SE). It develops to get rid of limits on attack instances, and the limits can direct cyber-attack anomalies. Zhang and Yang[13] developed a recognition technique besides FDIA for dynamic SE. The Kalman filter evaluates the state value from IEEE standard bus coordination. The long short-term memory (LSTM) model was employed to remove the successive annotations from conditions at manifold time-step. Furthermore, the model converts the attack recognition issue into a supervised learning issue and presents a deep neural network (DNN)-based recognition model. In[14], a NN-based technique is projected. The method converts the multi-dimension time-series measurement to their spectral and time-frequency graph representation. The developed model initially assumes the STFT and dual-channel convolutional neural network (2 C-CNN) to perfect the spectral-temporal links. Chen et al.[15] presented a DL recognition method based on the STFT approach. Firstly, low-quality data imitated by FDIA is detached utilizing a BDD detector. Next, the filtered measurement data is changed from the domain of time to the time-frequency domain utilizing STFT. Lastly, the outcomes of the STFT are input into a 2 C-CNN to remove the time-frequency domain feature for learning and training. In[16], a modified Red Fox Optimizer with a DL-enabled FDIA detection (MRFO-DLFDIAD) technique is offered in the atmosphere of CPPS. For recognition of FDIA, the proposed model employs a multi-head attention-based LSTM (MBA-LSTM) technique. The MRFO model is used in this research to enhance the recognition performance of the MBA-LSTM technique. Ahmadzadeh et al.[17] projected a data-driven structure to identify FDI assaults beside the power rule of PV combined PDS. Firstly, an attack-free model was modelled beside its power rule structure. In contrast, the grid measurement is directed to a central regulator, and the control signs are conveyed back to PV, which their local regulators employ. Afterwards, CNN's framework was developed to identify FDI assaults.

Chen et al.[18] presented a recognition structure of FDIA for PSSE based on GECCN, which utilizes topology data, edge feature, and node feature. Over the structure of a deep graph, the association of data is efficiently extracted to begin the connection between the assessed measurement value and the real state of power models. Furthermore, the edge-conditioned convolutional process permits processing datasets with dissimilar structures of graphs. Mukherjee, Chakraborty, and Ghosh[19] propose a DL model for real-time detection of data intrusions, efficiently detecting both structured and unstructured FDIA. Mukherjee et al.[20] introduce a novel real-time scheme for detecting FDIA through a DL-based state forecasting model, complemented by an innovative intrusion detection method utilizing the error covariance matrix. Optimized with carefully chosen hyperparameters, the architecture presents a scalable and effectual technique with minimal error margins. Mukherjee et al.[21] propose a scalable NN structure that efficiently anticipates operational states, emphasizing its capability to detect attacks in real-time. With optimized hyperparameters, this model attains superior state forecasting accuracy with minimal error compared to contemporary methods. Mahmoudnezhad et al.[22] developed a cyber-resilient hybrid DL model, namely SMDAE-GAN, that precisely forecasts electric load for both short-term and long-term horizons. The methodology incorporates a stacked multi-layer denoising autoencoder (SMDAE) for preprocessing and anomaly removal and a generative adversarial network (GAN) for load value forecasting. Zhou et al.[23] propose a dataset balancing model incorporating GAN-Gated Recurrent Units (GAN-GRU) with an FDIA detection model based on Transformer NNs. He et al.[24] present an A-BiTG model for detecting FDIA that incorporates a bidirectional temporal convolutional network and a bidirectional GRU with an attention mechanism. This approach improves feature extraction by utilizing past and future temporal data, illustrating greater accuracy than existing methods on the IEEE 14-bus and IEEE 118-bus systems. Radhoush et al.[25] introduce a novel deep NN (DNN) approach using real measurements. Wang, Zhou, and Ma[26] introduce the AT-GVAE framework for locational FDIA detection, which integrates variational autoencoders (VAE) and GANs. The framework employs dual VAE modules and multi-layer GRU to capture temporal correlations in measurement data, improving detection accuracy. Huang, Li, and Wang[27] present a deep reinforcement learning (DR) method that integrates state attention to improve state feature extraction. By incorporating an attention mechanism into the model-free DR learning method, the methodology concentrates on critical state features, enhancing the accuracy and speed of attack detection.

Lawal et al.[28] developed a robust machine learning (ML) model to reduce FDIA in dynamic line rating (DLR) systems by incorporating statistical data processing with a minimum redundancy maximum relevance (MR-MR) feature ranking and selection model. Tirulo et al.[29] introduce a novel unsupervised learning model for detecting FDIAs in smart grids by incorporating K-means clustering with the spectral residual method (KM-SR). Gupta et al.[30] present a methodology utilizing energy metering data through the AdaBoost ensemble model. The approach efficiently handles unbalanced data and is trained on malicious and non-malicious samples across five attack scenarios. Su et al.[31] presents DAMGAT, an interpretable DL method for detecting FDIAs. The model also utilizes a dual-attention mechanism in a multi-head graph attention network to efficiently capture correlations between attack characteristics and measurement data. Han et al.[32] proposes a novel FDIA localization detection method that utilizes graph data modeling and DL, which converts data into graph structures, designs specialized networks for different topologies, and utilizes a multi-graph mechanism. Liu and Dai[33] introduces a novel DL framework by integrating BERT and LSTM networks. By employing BERT's contextual encoding and LSTM's sequential processing, the model dynamically extracts features to generate embedding vectors that effectually detect malicious SQL queries. Li, Wang, and Lu[34] proposes a false data detection method for power grids by utilizing a gated graph NN (GGNN) model. An attention mechanism improves node representation by assigning weights to neighboring nodes. Kesici, Pal, and Yang[35] proposes a vertical federated learning (FL) framework, which enables sub-networks to collaboratively build an FDIA detection model by utilizing an attention module for spatial features on the edge side and a Bi-LSTM for temporal features on the server side. Alarfaj and Khan[36] presents a probabilistic NN (PNN) model to detect SQL injection attacks, optimizing the smoothing parameter with the BAT approach. Li, Hu, and Lu[37] introduces an FDIA detection methodology by using a Spatial-Temporal Transformer network that incorporates power grid topology and data. The model also captures global and local dependencies through self-attention and graph convolution. Sengan et al.[38] proposes a DL-based Cybertwin-improved LSTM-anomaly detection (DL-Cyberwin-Improved LSTM-AD) model.

Vu et al.[39] presents an efficient automated labeling approach. The method incorporates transfer learning (TL) with attention mechanisms to improve abnormal signal detection efficiency. Xia, He, and Yu[40] develops a graph convolutional attention network (GCAT) model, which employs graph topology and attention mechanisms to improve locational detection performance. Feng, Lao, and Chen[41] proposes the logit-normalized Bayesian ResNet (LNBRN) approach. This technique uses dropout for Bayesian inference and logit normalization to mitigate overconfidence. Phu et al.[42] introduces Graph CNN (GCNN) models. The technique starts with two classes namely suspicious and non-suspicious nodes and segments the network into varying security policies by utilizing distributed Ryu controllers in an SDN framework. Atheeq et al.[43] aims to develop a mechanism for identifying and attributing two-level ensemble attacks specifically targeting Industrial Control Systems (ICSs), utilizing a novel ensemble DL method integrated with decision trees and DNNs. Ding, Abdel-Basset, and Mohamed[44] introduces DeepAK-IoT, a DL method for detecting cyberattacks on IoT devices comprising three blocks. Bergies et al.[45] introduces an IoT architecture that integrates model predictive control (MPC) and DNNs techniques. Trajectory prediction and Homomorphic Encryption is also utilized for secure anomaly detection. Li, Ma, and Sun[46] presents an Adaptive DL (ADL) method that incorporates data preprocessing, NN pre-training, and classification utilizing high-dimensional features. The model optimizes architecture depending on traffic characteristics and improves classification by incorporating DL with conventional ML model. Ullah et al.[47] proposes an enhanced IDS for IoT security that employs multimodal big data and TL. It processes Packet Capture (PCAP) files, optimizes data with Spark, and incorporates text and texture features from word2vec and an attention-based Residual Network (ResNet) for attack classification. Varshini and Latha[48] explores the impact of attacks on WAC applications and evaluates detection utilizing model-based and data-driven methods. Oyinloye, Arowolo, and Prasad[49] provides an updated learning strategy for artificial NN (ANN) model. Wang et al.[50] proposes a deep semi-supervised numerical false data detection (DSS-NFDD) methodology by utilizing PageRank for dimensionality reduction, a semi-supervised DL framework, and a concentration loss function to mitigate noise and bias.

The existing studies emphasize diverse limitations in their methods of detecting FDIAs. For instance, some models may need help with scalability and generalizability to diverse conditions due to their dependence on tailored loss functions or specific transformations, which may present data loss. Others may need to perform better in scenarios with insufficient labelled data or when faced with noisy inputs, potentially affecting their accuracy. Furthermore, the complexity of incorporating various architectures can complicate training processes and limit practical implementation, while specific models may be sensitive to parameter tuning, affecting their reliability in dynamic environments. Some unsupervised and ensemble models may need help detecting rare attack patterns, resulting in missed anomalies, specifically when dealing with high-dimensional or imbalanced datasets. Few studies include interpretable DL models employing attention mechanisms, frameworks integrating diverse NNs for an improved feature extraction, and collaborative detection methods in FL. Threats namely scalability, performance degradation under noise, and potential biases in classification strategies are noted, affecting generalizability and real-time applicability. There is a requirement for more robust and adaptable detection methods for FDIAs that can efficiently handle dynamic and growing attack patterns in real-time. Existing models struggle with scalability, generalizability, and performance under varying conditions, accentuating the necessity for innovative models incorporating diverse data sources and utilizing advanced ML models to enhance detection accuracy.

Proposed methodology

This study presents a novel ARDL-FDIAR approach. The primary goal of the ARDL-FDIAR approach is to boost security via the FDIA detection process. To accomplish that, the ARDL-FDIAR technique has four stages: z-score normalization, MLOA-based selection feature, IDBN-based FDIA detection, and COA-based parameter selection. Figure 1 illustrates the workflow of the ARDL-FDIAR methodology.

Z-score normalization

At the primary stage, the ARDL-FDIAR technique takes place, and Z-score normalization is used for scaling the input data[51]. This model is chosen for data preprocessing due to its efficiency in standardizing features. It transforms them into a common scale with a mean of zero and a standard deviation (SD) of one. This technique is particularly advantageous when dealing with datasets that exhibit varying scales and distributions, as it reduces the influence of outliers and improves the stability of model training. Unlike min-max scaling, which can compress values into a limited range, Z-score normalization conserves the distribution shape, making it appropriate for algorithms that assume normally distributed data. Moreover, this methodology facilitates faster convergence during optimization, as it assists in maintaining a consistent range of input values, ultimately enhancing the performance of the technique. Its simplicity and efficiency in improving interpretability and comparability across features justify its widespread use in ML applications.

Z-score normalization (standardization) is a statistical method to normalize the data features. This technique transforms the information into a mean (μ) of 0 and an SD (σ) of 1. The equation is given below:

$$z = \frac{(x - \mu)}{\sigma} \tag{1}$$

Where x refers to the original value, μ denotes the mean of the feature, and σ shows the SD.

Standardizing features ensures that all contribute equally to the analysis and prevents features with large scales from dominating those with small scales. This is especially helpful in ML techniques, as it improves the model's performance and convergence. Furthermore, it makes the data more related across dissimilar features and facilitates better analysis.

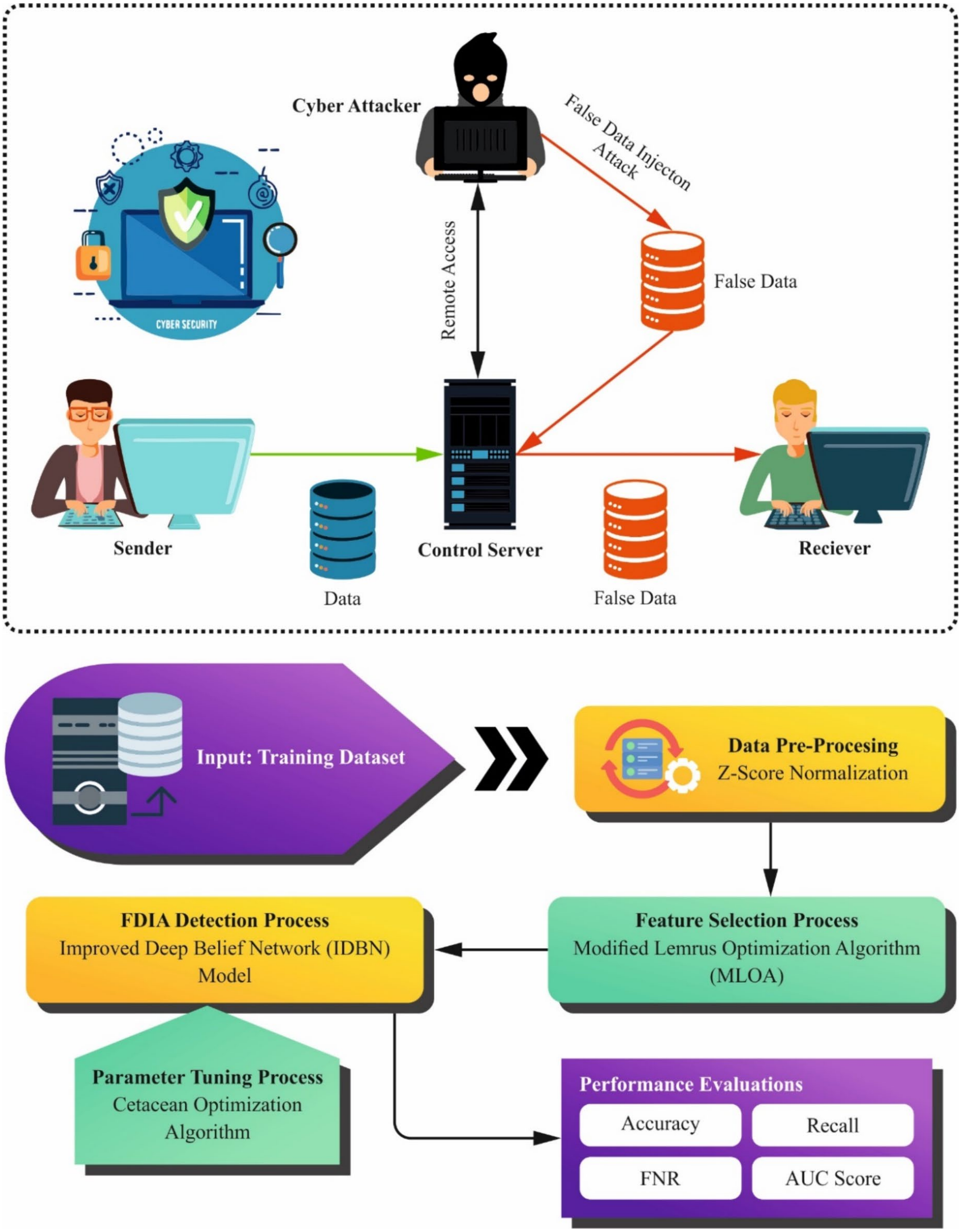

Fig. 1. Workflow of ARDL-FDIAR technique.

MLOA-based feature selection

Next, the attribute reduction process gets invoked using MLOA to choose optimal feature sets[52]. This model was selected for feature selection due to its effectual balance between exploration and exploitation, which was motivated by the foraging behaviour of lemurs. This methodology outperforms at navigating high-dimensional feature spaces, assisting in detecting the most relevant features while minimizing redundancy and noise. MLOA's adaptive mechanisms allow it to alter its search strategy dynamically, improving its capability to escape local optima and find global solutions. Compared to conventional approaches, MLOA can enhance model accuracy by choosing a more compact and informative subset of features, ultimately mitigating overfitting. Its robustness in handling diverse datasets makes it a valuable choice for optimizing ML model performance, specifically in complex environments. Figure 2 illustrates the workflow of the MLOA model.

LOA was presented to resolve global optimizer problems and is established depending on the stimulus from Lemurs' social behaviour. Lemurs are prosimian primates, and unlike monkeys and apes, they are classified separately within the primate group. They typically live in groups known as troops. LOA mainly utilizes two kinds of locomotive behaviour revealed by Lemurs, such as leap-up and dance-hup. The previous is the stimulus

Fig. 2. Workflow of MLOA technique.

for the exploitation stage as a local search, and the final is the stimulus for the exploration stage as a global search. The lemur's position has assumed that candidate performances will be upgraded in all the iterations depending on the fitness function (FF). The lemur population is usually determined as exposed in Eq. (2).

$$T = \begin{bmatrix} L_1^1 & \ldots & L_1^d \\ \vdots & \ddots & \vdots \\ L_s^1 & \ldots & L_s^d \end{bmatrix} \tag{2}$$

The population matrix size is $s * d$, whereas s implies the candidate solution counts are equivalent to lemur counts, and d demonstrates the decision variable counts. As exposed in Eq. (3), the lemur positions can usually initialize randomly.

$$L_i^j = rand\,() \times \,(ub_j - lb_j) + 1b)\,\forall\, i \in \,(1,\,2,\,\ldots\,,\,n)\,\wedge\,\forall\, j \in\,(1{,}2,\,\ldots\,,\,d) \tag{3}$$

At this point, it signifies the random number between zero and one; upper and lower boundaries are represented by ub_j and lb_j, respectively. During every iteration, the global best lemur and best nearby lemur are recognized depending on FF and lemur positions are upgraded based on the next Eq. (4):

$$L_{i+1}^j = \begin{cases} L_i^j + abs(L_i^j - gbest^j) * (rand - 0.5) * 2 \\ if rand \geq \ FRR \\ L_i^j + abs(L_i^j - nbest^j) * (rand - 0.5) * 2 \\ if\ rand < FRR \end{cases} \tag{4}$$

Whereas i and j denote the iteration numbers and decision variable, correspondingly, $gbest$ implies the position of the global best Lemur, and $nbest$ denotes the position of the best nearby lemur; L refers to the lemur location, $rand$ denotes the random number among zero and one; $Free\ Risk\ Rate\ (FRR)$ denotes the risk level of every lemur from the troop, and this co-efficient has been measured by Eq. (5).

$$FRR\ = FRR\,(HRR) - Crnt\_Iter \times \left(\frac{HRR - LRR}{Max\_Iter}\right) \tag{5}$$

In Eq. (5), HRR signifies the High-Risk Rate, and LRR demonstrates the Low-Risk Rate; they are constant existing rates. $Crnt\_Iter$ defines the present iteration, and Max_Iter implies the maximal iteration counts.

The two essential modifications are presented in the LOA for the progress of MLOA. Primarily, this formula utilized for updating the position of Lemur's defined in Eq. (4) is adapted as follows (6):

$$L_{i+1}^j = \begin{cases} L_i^j + r1 * sin\left(abs\left(L_i^j - gbest^j\right)\right) \\ * (rand - 0.5) * 2if\ FRR \geq \ 0.5 \\ L_i^j + r1 * cos\left(abs\left(L_i^j - nbest^j\right)\right) \\ * (rand - 0.5) * 2if\ FRR < 0.5 \end{cases} \tag{6}$$

Equation (6) is stimulated from the Sine Cosine Algorithm (SCA), and the procedure of Sine & Cosine functions from an optimizer method enhances the swarm's exploration and exploitation proficiency. The rate of r1 is measured as Eq. (7).

$$r1 = a - Crnt\_Iter \times \left(\frac{a}{\mathrm{Max}_{Iter}}\right) \tag{7}$$

In Eq. (7), $Crnt\_Iter$ signifies the present iteration, and Max_Iter implies the maximal iteration counts; a refers to the constant, which is assumed to be 3. In Eq. (6), the FRR is utilized for selecting exploration or exploitation phases from all the iterations by analyzing it with a constant rather than the random value employed in the new LOA. Introducing this constant is responsible for optimal convergence, and experimental assessments have determined that the optimal rate of this constant is 0.5.

The second alteration presented in MLOA, if related to the new LOA, is the established upgrading formula to update the rate of the worst lemur from all the iterations. The lemur with the minimum fitness value has been recognized, and the position of worse lemurs is upgraded utilizing Eq. (8).

$$L_{worst,i} = L_{\min}\ + (L_{\max}\ - L_{\min}\ ) * rand \tag{8}$$

whereas $L_{\max}$ and $L_{\min}$ signify the maximal and minimal lemur locations, respectively, they are calculated depending on the primary lemur population; $rand$ denotes the random number between zero and one.

The FF assumes that the classifier accuracy and the selection feature count. It exploits the classifier accuracy and decreases the set dimension of chosen features. Hence, the succeeding FF is organized for calculating distinct solutions, as expressed in Eq. (9).

$$Fitness = \alpha\ *\ ErrorRate + (1 - \alpha\ ) * \frac{\#SF}{\#All\_F} \tag{9}$$

Whereas $ErrorRate$ signifies the classifier error value employing the chosen features. $ErrorRate$ has been evaluated as the percentage of improper classification to the count of classifiers made, exemplified as a value between 0 and 1. $\#SF$ defines the chosen feature counts, and $\#All\_F$ implies the entire feature counts within the novel database. α has been employed for adjusting the effect of classifier quality and subset length. In the research, a value of 0.9 is assigned to α.

FDIA detection using IDBN model

The FDIA detection process is performed at this stage using the IDBN model[53]. This model is chosen because it can capture intrinsic, hierarchical feature representations from high-dimensional data. Unlike conventional methods, the IDBN model utilizes diverse layers of stochastic, latent variables, enabling it to learn complex patterns and dependencies within the data indicative of attacks. Its generative nature allows for effectual unsupervised pre-training, improving performance even in scenarios with limited labelled data. Moreover, the model's architecture facilitates fine-tuning, making it adaptable to specific contexts and attack types. This flexibility, integrated with its superior learning capacity, positions the IDBN as a powerful tool for precisely detecting FDIAs in dynamic environments. Figure 3 depicts the structure of the DBN model.

DBN is composed of several RBM stacks and classifications at the top level. Every RBM encompasses a visible layer (VL) and a hidden layer (HL). Weights have linked the VL and HL, and the neurons from the similar layer are independent.

DBN continuously removes the significant features of the preceding resultant layer with RBM, which recognizes the stepwise training from the bottom to the top. An optimum mapping among every RBM was acquired during the adjustment model of the entire DBN infrastructure, along with training. It also requires utilizing the BP technique to slowly propagate the error among the resultant values and predictable rates from the top to bottom layers.

In the training method, the neuron layers from the VL of RMB are represented as $v=\{v_1,\ v_2,\ \cdots,\ v_j\}$, and the neuron layers from the HL are defined as $h=\{h_1,\ h_2,\ \cdots,\ h_j\}$. The proficiency function of RBM is written as:

$$E\left(v,\ h\right)=-\sum\nolimits_{i=1}^{n}a_iv_i-\sum\nolimits_{j=1}^{m}b_ih_i-\sum\nolimits_{i=1}^{n}\sum\nolimits_{j=1}^{m}w_{ij}v_ih_i \tag{10}$$

Whereas a suggests the offset vector of VL; b denotes the offset vector of HL; w demonstrates the weighted matrix connecting the VL and HL.

If every parameter is defined, Eq. (10) is represented by the joint probability distribution function among the VL as well as HL, as expressed in

$$P\left(v,\ h\right)=\frac{1}{\sum\nolimits_{\mathrm{i}=1}^{m}\sum\nolimits_{\mathrm{j}=1}^{n}e^{-E(v,h)}}e^{E(v,h)} \tag{11}$$

Since there is no correlation among neurons from all the layers, it is essential to utilize the activation function to measure whether neurons are activated. The generally utilized activation functions in DL comprise Tanh, Sigmoid, ReLU, and so on. If the sigmoid and Tanh functions method has positive and negative infinity, the gradient disappears, and the weight upgrade speed reduces. The ReLU function resolves the problem of vanishing gradients; then, the derivative remains 0 at negative rates. It is inclined to neuronal necrosis. The hard swish function not only ensures the benefits of the ReLU function but also overcomes the computational mistakes. It is smoother and takes a faster rate of convergence speed. The hard swish function is defined as:

$$Hard\ swish\left(x\right)=x\frac{ReLU6\left(x+3\right)}{6} \tag{12}$$

$$ReLU6=min\left(ReLU,\ 6\right) \tag{13}$$

The conditional distribution probability for VL and HL is:

$$P\left(v_i=1|v\right)=Hard\ swish\ \left(b_j+\sum\nolimits_{j=1}^{n}v_jw_{ij}\right) \tag{14}$$

$$P\left(v_i=1|h\right)=Hard\ swish\ \left(a_j+\sum\nolimits_{j=1}^{n}h_jw_{ij}\right) \tag{15}$$

A new Sigmoid function is exchanged with the Hard swish function. It resolves the problem of the gradient disappearing and the weighted upgrade speed diminishing. So, the accuracy of DBN is enhanced.

Hyperparameter tuning using COA

Finally, the performance of the IDBN model has been improved by the design of a COA-based hyperparameter tuning process[54]. This model is chosen for hyperparameter tuning due to its unique capability to replicate cetaceans' social behaviours and hunting strategies, which improves exploration and exploitation in the search space. This bio-inspired approach allows for efficient convergence toward optimal hyperparameters by employing diverse search strategies that can adapt to intrinsic landscapes typical in ML techniques. Unlike conventional tuning models, COA can effectively navigate high-dimensional spaces and is less likely to become trapped in local optima. Its robustness in handling nonlinear relationships between hyperparameters enhances the model's

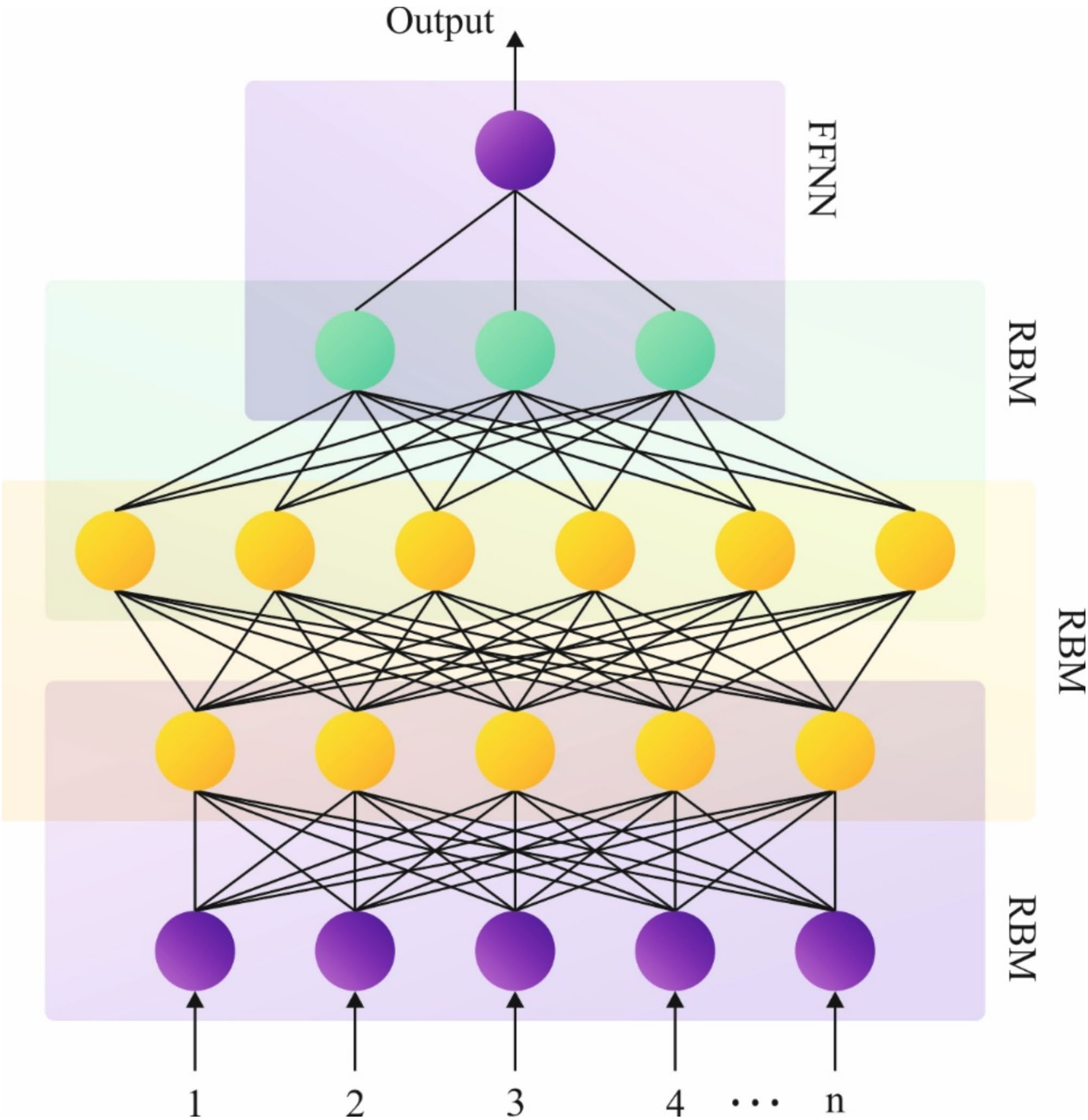

Fig. 3. Structure of the DBN technique.

performance. Moreover, the flexibility of the COA model makes it appropriate for several optimization issues, making it a versatile choice for improving the accuracy and effectiveness of ML models. Figure 4 demonstrates the workflow of the COA approach.

The COA is a population-based stochastic optimizer approach stimulated by nature that is just produced and deploys a set of searching agents for efficiently addressing optimizer problems. Drawing inspiration from humpback cetaceans, the COA's bubble-net hunting approach comprises three major elements: bubble net attacking, encircling prey, and surveying nearby the optimum prey. The cetaceans' bubble-net foraging principle involves generating a spiral bubble net once recognizing prey and moving upstream to take it. This predation behaviour is revealed in 3 phases: hunting prey, surrounding prey, and bubble-net attack. Related to cetaceans encompassing fishes under hunting, the COA agents fine-tune their places to determine the optimum choice for histogram augmentation.

$$\mathfrak{x}(t+1) = \begin{cases} \mathfrak{x}^{*}(t) - u \times \left|\mathbb{C} \times \mathfrak{x}*(t) - \mathfrak{x}(t)\right| & if\ < 0.5 \\ \left|\mathbb{C} \times \mathfrak{x}*(t) - \mathfrak{x}(t)\right| \times e^{b \cdot \updownarrow} \cdot cos(2\pi t) + \mathfrak{x}*(t) & if\ \geq 0.5 \end{cases} \quad (16)$$

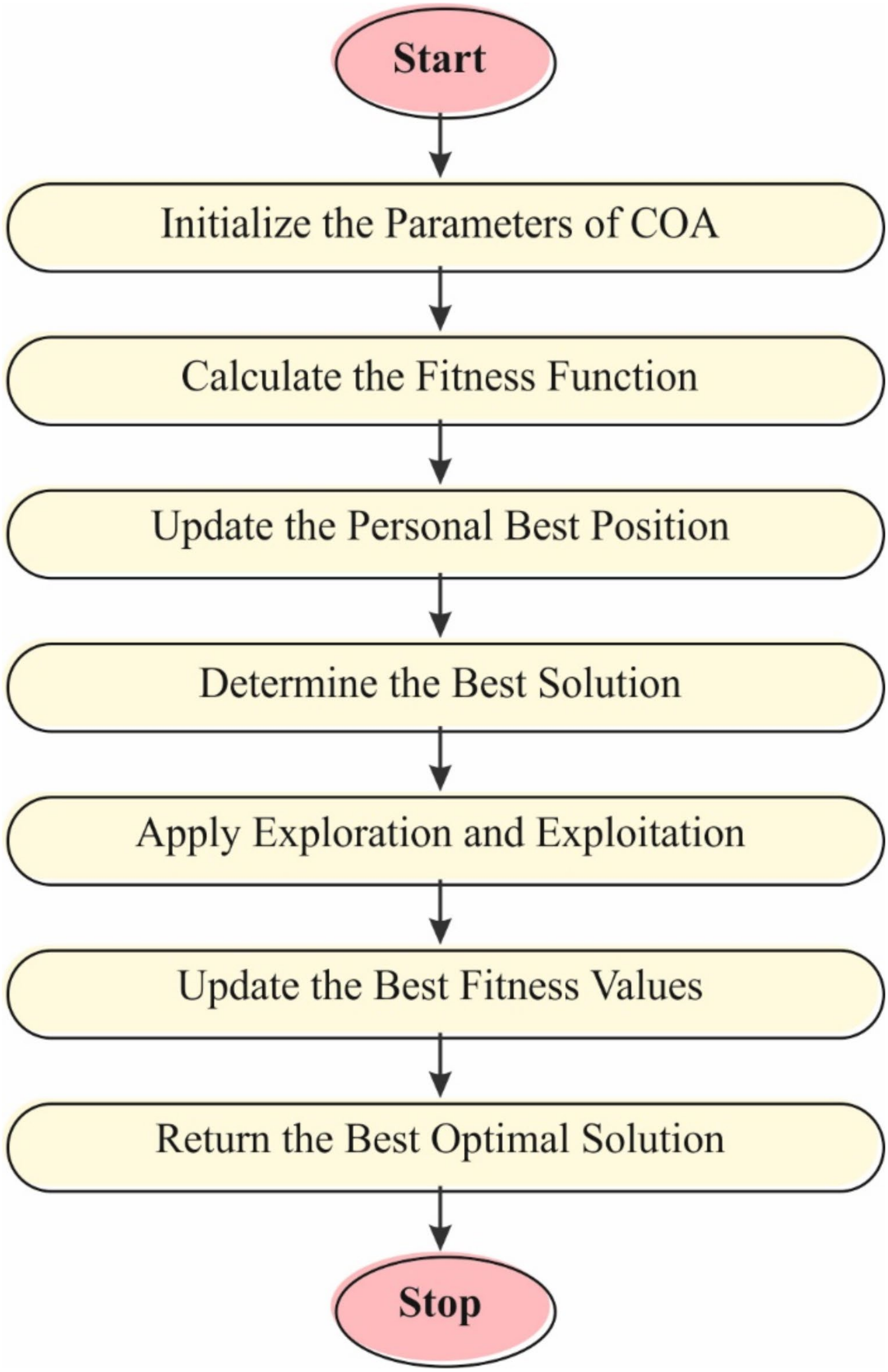

Fig. 4. Workflow of the COA method.

Once t refers to the time or iteration index, $\mathfrak{x}$ denotes the vector comprising every place of cetaceans, and $\mathfrak{x}*$ signifies the optimum solution; a coefficient vector with a linear reduction from [2 – 0] through the repetition counts is $= 2a.\,(r - a)\,;\mathfrak{C} = 2.r$; The variables r and $\lfloor$ identify the procedure of logarithmic spiral, correspondingly, depending on the input image. During this case, $\lfloor$ is fixed to 1, but r signifies the random vector with values between zero and one. In Eq. (16), the probability 5 $\mathcal{P}$ is 50% and 50%. $\updownarrow$ defines the random integer among [-1,1], which can employed for swapping upgrade the place of cetaceans; it points out that, with an equivalent probability, cetaceans elect both selections at random throughout the optimizer model. The random rate for $\sqcap$ from the bubble-net stage is $(-1,1)$, but the random rate of vector A from the searching stage might be more or less than 1.

Equation (17) demonstrates the searching mechanisms.

$$\mathfrak{x}(t+1) = \mathfrak{x}_r - \mu \times \left|\mathbb{C} \times \mathfrak{x}(t) - \mathfrak{x}(t)\right| \tag{17}$$

The random search model, featuring a $|u|$ rate exceeding one, accentuates the exploration method and requires the COA approach for a widespread global search. Primarily, random solutions are created at the beginning of the COA search. Next, these solutions endure iterative upgrades over the model. Until a set of maximal iteration counts is accomplished, the search continues.

The COA develops an FF to accomplish greater classifier efficiency. It resolves the positive integer to signify the optimum efficacy of candidate results. During this effort, the reduction of the classifier error value is supposed to be FF, as defined in Eq. (18).

$$fitness\,(x_i) = ClassifierErrorRate\,(x_i) = \frac{No.\ of\ misclassified\ samples}{Total\ no.\ of\ samples} \times 100 \tag{18}$$

Experimental validation

The experimental validation analysis of ARDL-FDIAR methodology is examined using the IEEE14 standard system and the IEEE39 standard system[16]. The suggested technique is simulated using the Python 3.6.5 tool on a PC with an i5-8600k, 250GB SSD, GeForce 1050Ti 4GB, 16GB RAM, and 1 TB HDD. The parameter settings are provided: learning rate: 0.01, activation: ReLU, epoch count: 50, dropout: 0.5, and batch size: 5.

Table 1; Fig. 5 demonstrate the $accu_y$ outcome of the ARDL-FDIAR technique with various attack intensities. On robust attack intensity, the ARDL-FDIAR technique has resulted in $accu_y$ of 97.16% in power, 94.34% in voltage amplitude, and 93.41% in voltage phase-angle. In addition, on mid-attack intensity, the ARDL-FDIAR approach has resulted in an $accu_y$ of 93.24% in power, 94.17% in voltage amplitude, and 92.20% in voltage phase-angle. Finally, on weak attack intensity, the ARDL-FDIAR approach has resulted in an $accu_y$ of 90.22% in power, 91.70% in voltage amplitude, and 89.88% in voltage phase-angle.

Table 2 depicts the detection outcome of the ARDL-FDIAR technique with other approaches under the IEEE14 standard system[16,55,56]. Figure 6 illustrates the brief $accu_y$ analysis of the ARDL-FDIAR technique with existing models under two phases. The experimental values imply that the ARDL-FDIAR technique has maximum performance. For instance, in the training phase, the ARDL-FDIAR technique has accomplished an enhanced $accu_y$ of 97.91%, but the DNN, RF, LSTM, XGBoost, Semi-Supervised Generative Adversarial Network with Convolutional Neural Network (SGAN + CNN), Semi-Supervised Generative Adversarial Network with Graph Convolutional Network (SGAN + GCN), Semi-Supervised Generative Adversarial Network with Graph Attention Based Generation Layer (SGAN + GAL), Semi-Supervised Multi-Label Adversarial Network (SMAN), MRFO-DLFDIAD, Support Vector Machine (SVM), CNN, Margin Setting Algorithm (MSA), C-DBN, and SVM with Gentle Adaboost (SVM-GAB) approaches have a lesser $accu_y$ of 96.51%, 93.17%, 94.59%, 92.94%, 95.76%, 96.37%, 90.04%, 92.04%, 97.24%, 90.48%, 89.85%, 90.68%, 92.10%, and 91.04%, respectively. Moreover, in the testing phase, the ARDL-FDIAR technique achieved a superior $accu_y$ of 98.58%. Still, the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches have reduced $accu_y$ of 95.64%, 94.17%, 93.90%, 95.86%, 96.62%, 95.69%, 94.95%, 94.80%, 97.95%, 91.48%, 92.40%, 90.46%, 93.99%, and 93.95%, correspondingly.

Figure 7 exemplifies the detailed $reca_l$ examination of the ARDL-FDIAR technique with existing models under two phases. The outcome values stated that the ARDL-FDIAR technique has higher performance. For instance, in the training phase, the ARDL-FDIAR technique has reached a greater $reca_l$ of 97.71%, but the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches have a lower $reca_l$ of 95.60%, 93.30%, 94.16%, 94.35%, 93.05%, 91.49%, 92.26%, 95.17%, 97.20%, 92.09%, 93.30%, 90.72%, 92.89%, and 92.31%, correspondingly. Additionally, in the testing phase, the ARDL-FDIAR methodology obtained a maximal $reca_l$ of 98.57%. In contrast, the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches have minimal $reca_l$ of 92.63%, 94.41%, 90.25%, 95.57%, 91.56%, 90.60%, 96.13%, 95.75%, 97.85%, 90.85%, 93.15%, 89.63%, 93.11%, and 91.68%, correspondingly.

Figure 8 demonstrates the brief FNR investigation of the ARDL-FDIAR method with existing methodologies under two phases. The experimental values imply that the ARDL-FDIAR method performs better. For instance, in the training phase, the ARDL-FDIAR method has reached a minimal FNR of 2.28, whereas the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches have higher FNR of 8.85%, 8.12%, 9.32%, 10.00%, 9.23%, 8.65%, 8.15%,

Accuracy (%)			
Attack Intensity	**Power**	**Voltage Amplitude**	**Voltage Phase-Angle**
Strong	97.13	94.34	93.41
Mid	93.24	94.17	92.20
Weak	90.22	91.70	89.88

Table 1. $Accu_y$ outcome of ARDL-FDIAR technique with various attack intensity.

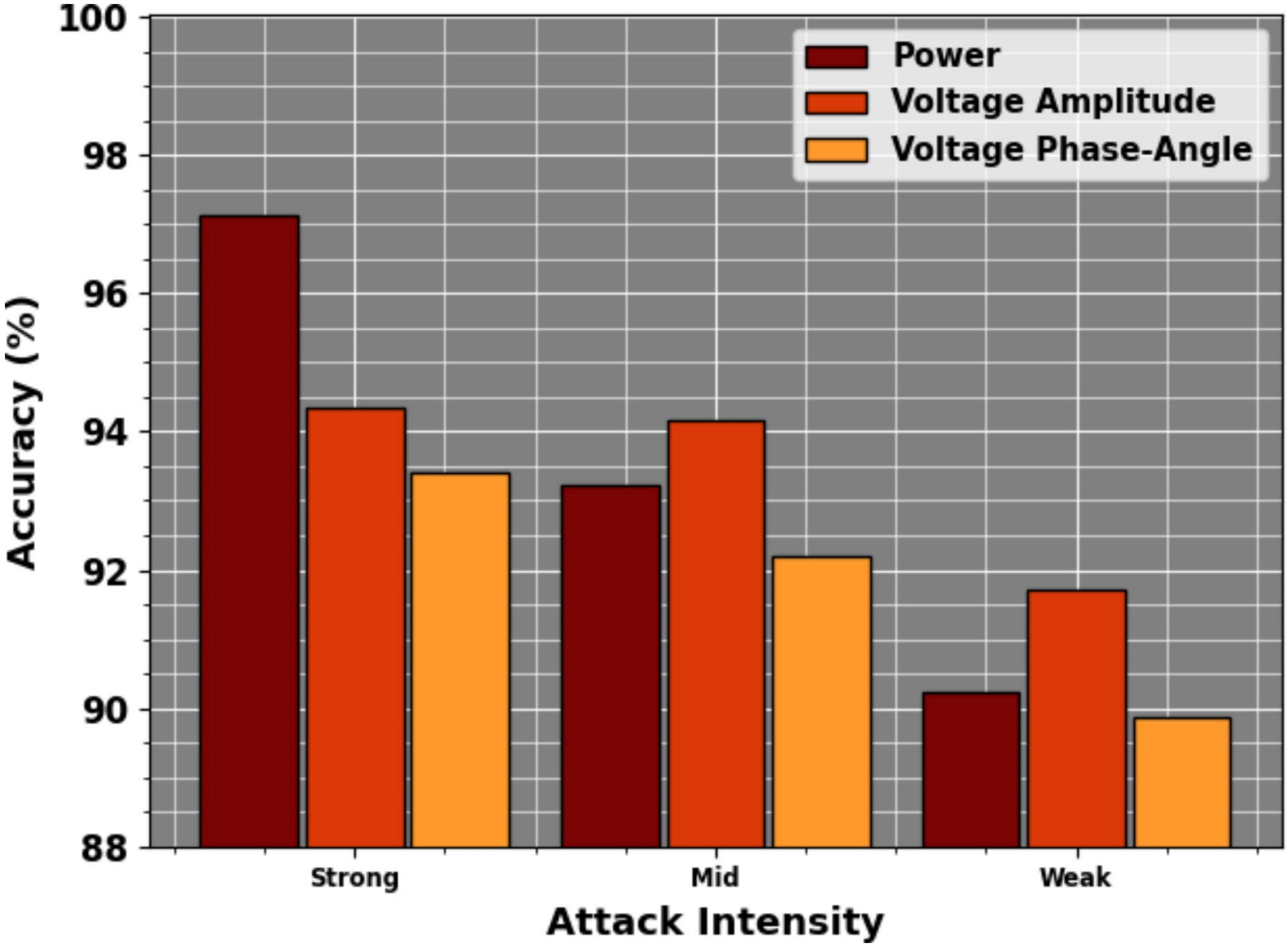

Fig. 5. $Accu_y$ outcome of ARDL-FDIAR model with various attack intensity.

8.09%, 3.64%, 8.75%, 7.53%, 10.02%, 7.69%, and 8.32%, respectively. Besides the testing phase, the ARDL-FDIAR approaches achieved a lesser FNR of 1.56, whereas the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches have higher FNR of 7.63%, 6.46%, 6.19%, 6.67%, 7.84%, 7.47%, 7.83%, 6.32%, 2.93%, 9.75%, 7.66%, 11.08%, 7.65%, and 9.22%, correspondingly.

Figure 9 shows the brief AUC_{score} analysis of the ARDL-FDIAR model with existing approaches under two phases. The experimental outcomes inferred imply that the ARDL-FDIAR model performs better. For instance, in the training phase, the ARDL-FDIAR approach attained a higher AUC_{score} of 98.55%, whereas the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches had smaller AUC_{score} of 92.22%, 92.54%, 94.78%, 90.17%, 91.54%, 95.27%, 90.94%, 95.23%, 97.97%, 92.17%, 91.19%, 90.05%, 93.86%, and 92.51%, correspondingly. Additionally, in the testing phase, the ARDL-FDIAR model has accomplished an enhanced AUC_{score} of 97.90%. Still, the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches have lesser AUC_{score} of 92.26%, 90.49%, 93.81%, 93.79%, 92.72%, 96.38%, 92.37%, 93.77%, 97.21%, 91.73%, 91.18%, 91.85%, 90.30%, and 90.59%, correspondingly.

In Fig. 10, the training and validation accuracy outcomes of the ARDL-FDIAR approach under the IEEE14 standard system are revealed. The accuracy rates are calculated throughout 0–25 epochs. The outcome highlighted that the training and validation accuracy values demonstrated an increasing tendency, which reported the proficiency of the ARDL-FDIAR approach with enhanced performance over many iterations. Furthermore, the training and validation accuracy remain closer over the epochs, which point out low minimal overfitting and exhibits superior outcomes of the ARDL-FDIAR method, guaranteeing consistent prediction on unseen samples.

In Fig. 11, the training and validation loss graph of the ARDL-FDIAR approach under the IEEE14 standard system is exhibited. The loss values are calculated over an interval of 0–25 epochs. The training and validation accuracy rates exemplify a decreasing tendency, reporting the proficiency that notified the ability of the ARDL-FDIAR approach to balance a trade-off between data fitting and generalization. The ensuing reduction in loss rates additionally guarantees the better performance of the ARDL-FDIAR methodology and tunes the prediction outcomes over time.

IEEE14 Standard System				
Methods	$Accu_y$	$Reca_l$	FNR	AUC Score
Training Phase				
ARDL-FDIAR	97.91	97.71	2.28	98.55
DNN	96.51	95.60	8.85	92.22
RF	93.17	93.30	8.12	92.54
LSTM	94.59	94.16	9.32	94.78
XGBoost	92.94	94.35	10.00	90.17
SGAN + CNN	95.76	93.05	9.23	91.54
SGAN + GCN	96.37	91.49	8.65	95.27
SGAN + GAL	90.04	92.26	8.15	90.94
SMAN	92.04	95.17	8.09	95.23
MRFO-DLFDIAD	97.24	97.20	3.64	97.97
SVM	90.48	92.09	8.75	92.17
CNN	89.85	93.30	7.53	91.19
MSA	90.68	90.72	10.02	90.05
C-DBN	92.10	92.89	7.69	93.86
SVM-GAB	91.04	92.31	8.32	92.51
Testing Phase				
ARDL-FDIAR	98.58	98.57	1.56	97.90
DNN	95.64	92.63	7.63	92.26
RF	94.17	94.41	6.46	90.49
LSTM	93.90	90.25	6.19	93.81
XGBoost	95.86	95.57	6.67	93.79
SGAN + CNN	96.62	91.56	7.84	92.72
SGAN + GCN	95.69	90.60	7.47	96.38
SGAN + GAL	94.95	96.13	7.83	92.37
SMAN	94.80	95.75	6.32	93.77
MRFO-DLFDIAD	97.95	97.85	2.93	97.21
SVM	91.48	90.81	9.75	91.73
CNN	92.40	93.15	7.66	91.18
MSA	90.46	89.63	11.08	91.85
C-DBN	93.99	93.11	7.65	90.30
SVM-GAB	93.95	91.68	9.22	90.59

Table 2. Comparative outcome of ARDL-FDIAR technique with other models under IEEE14 standard system[16,55,56].

Table 3; Fig. 12 illustrates the computational time (CT) analysis of the ARDL-FDIAR technique under IEEE14 standard system. The methods and their corresponding times are as follows: ARDL-FDIAR takes 6.47 s, while DNN requires 15.92 s. The RF method has a time of 15.25 s, and LSTM operates at 12.03 s. XGBoost performs in 9.60 s, whereas SGAN combined with CNN takes 10.95 s, and SGAN with GCN requires 12.34 s. SGAN with GAL operates in 10.68 s, while SMAN is the fastest at 9.08 s. MRFO-DLFDIAD takes 13.53 s, SVM requires 12.67 s, and CNN has a longer time of 18.55 s. MSA takes 17.09 s, C-DBN operates at 11.13 s, and SVMGAB has the longest time of 16.25 s.

Table 4 represents the detection outcome of ARDL-FDIAR methodology with other methods under the IEEE39 standard system.

Figure 13 demonstrates the detailed $accu_y$ outcome of the ARDL-FDIAR approach with existing models under two phases. The outcome values inferred that the ARDL-FDIAR approach has maximum performance. For instance, in the training phase, the ARDL-FDIAR methodology has attained a higher $accu_y$ of 98.09%, but the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches have a lesser $accu_y$ of 91.79%, 93.07%, 93.58%, 96.03%, 96.81%, 92.80%, 91.63%, 91.94%, 97.57%, 90.82%, 90.10%, 90.97%, 92.38%, and 91.35%, correspondingly. Also, in the testing phase, the ARDL-FDIAR methodology attained a greater $accu_y$ of 97.73%. In contrast, the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches have reduced $accu_y$ of 93.49%, 94.26%, 96.40%, 96.63%, 91.91%, 92.28%, 96.96%, 92.57%, 97.01%, 89.67%, 93.82%, 91.58%, 90.89%, and 89.91%, correspondingly.

Figure 14 demonstrates the brief $reca_l$ analysis of the ARDL-FDIAR technique with existing models under two phases. The experimental values inferred that the ARDL-FDIAR technique has maximum performance. For instance, in the training phase, the ARDL-FDIAR technique has accomplished a greater $reca_l$ of 98.27%, but the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN,

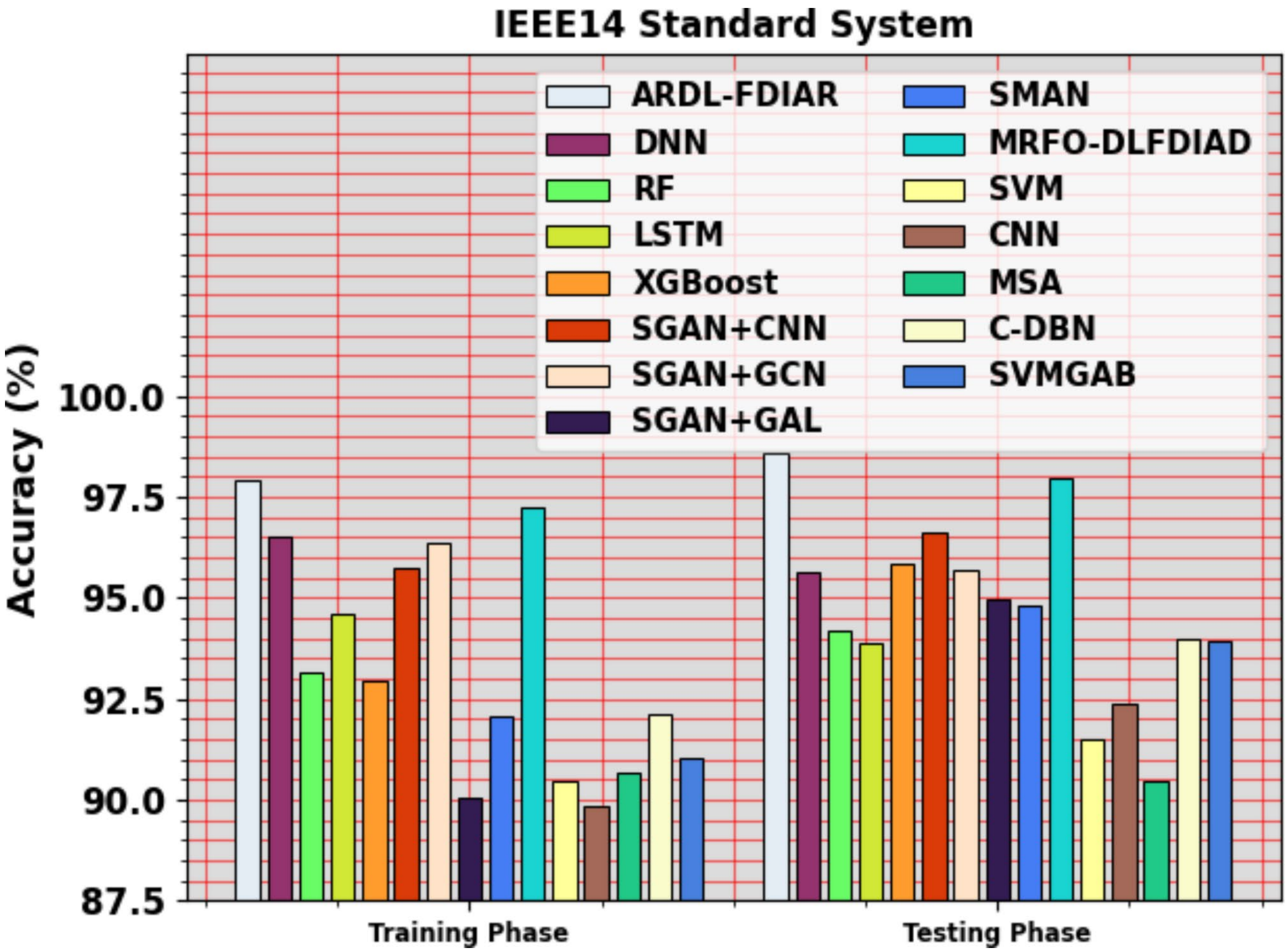

Fig. 6. $Accu_y$ outcome of ARDL-FDIAR technique under IEEE14 standard system.

MSA, C-DBN, and SVM-GAB methodologies have a lower $reca_l$ of 95.29%, 92.27%, 91.21%, 91.83%, 95.34%, 92.69%, 91.23%, 95.62%, 97.67%, 92.36%, 93.67%, 91.08%, 93.19%, and 92.63%, respectively. Furthermore, the ARDL-FDIAR technique has attained a superior $reca_l$ of 98.00% in the testing phase. In contrast, the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB methodologies have smaller $reca_l$ of 94.56%, 96.59%, 91.80%, 92.86%, 96.67%, 95.75%, 92.91%, 94.90%, 97.23%, 93.37%, 90.10%, 94.00%, 89.79%, and 92.20%, correspondingly.

Figure 15 illustrates the brief FNR analysis of the ARDL-FDIAR approach with existing models under two phases. The outcome stated that the ARDL-FDIAR approach performed better. For instance, in the training phase, the ARDL-FDIAR method has attained a reduced FNR of 2.58, whereas the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches have enhanced FNR of 7.23%, 5.79%, 5.44%, 5.95%, 7.23%, 5.09%, 5.20%, 5.69%, 3.98, 9.19, 7.89, 10.41, 7.99, and 8.66, correspondingly. Likewise, in the testing phase, the ARDL-FDIAR method has reached a lower FNR of 2.11 while the MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches have superior FNR of 6.38%, 5.78%, 6.80%, 5.41%, 7.00%, 6.23%, 5.81%, 5.35%, 3.36, 7.41, 10.69, 6.72, 10.90, and 8.59, correspondingly.

Figure 16 illustrates the brief AUC_{score} analysis of the ARDL-FDIAR technique with existing models under two phases. The outcome values implied that the ARDL-FDIAR technique also has maximal performances. For instance, in the training phase, the ARDL-FDIAR technique attained a superior AUC_{score} of 99.18%, whereas the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches attained decreased AUC_{score} of 93.19%, 95.88%, 92.58%, 95.23%, 93.00%, 97.12%, 92.73%, 96.89%, 98.38%, 92.47%, 91.58%, 90.37%, 94.12%, and 92.92%, respectively. Moreover, in the testing phase, the ARDL-FDIAR methodology reached a superior AUC_{score} of 98.35%. Still, the DNN, RF, LSTM, XGBoost, SGAN + CNN, SGAN + GCN, SGAN + GAL, SMAN, MRFO-DLFDIAD, SVM, CNN, MSA, C-DBN, and SVM-GAB approaches have a lower AUC_{score} of 96.47%, 91.83%, 94.95%, 95.01%, 96.33%, 95.68%, 95.48%, 94.46%, 97.57%, 92.88%, 92.97%, 90.43%, 92.08%, and 90.36%, correspondingly.

In Fig. 17, the training and validation accuracy results of the ARDL-FDIAR approach under the IEEE39 standard system are demonstrated. The accuracy rates are calculated from intervals of 0–25 epochs. The outcome showed that the training and validation accuracy values exhibit a rising tendency, which reported the ability of

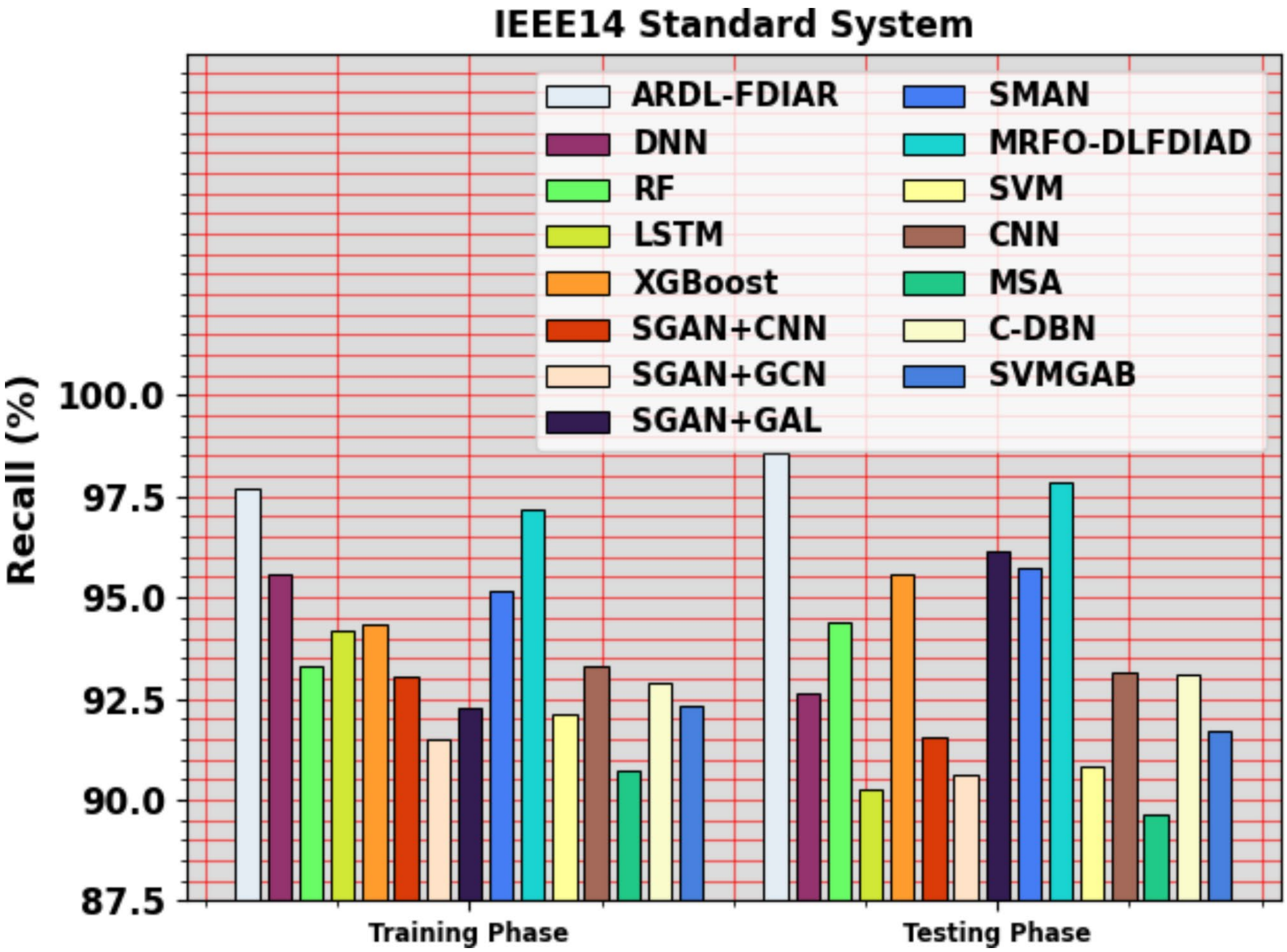

Fig. 7. $Reca_l$ outcome of ARDL-FDIAR technique under IEEE14 standard system.

the ARDL-FDIAR approach to improve performance under numerous iterations. Furthermore, the training and validation accuracy remain closer over the epochs, indicating low minimal overfitting and enhancing the ARDL-FDIAR approach's performance, promising consistent prediction on unseen instances.

Figure 18 depicts the training and validation loss graph of the ARDL-FDIAR technique under the IEEE39 standard system. The loss rates are calculated throughout 0–25 epochs. The training and validation accuracy values illustrate a decreasing tendency, notifying the competency that notified the capability of the ARDL-FDIAR technique in balancing a trade-off between data fitting and generalization. The continual decrease in loss values ensures the ARDL-FDIAR approach's higher performance and tunes the prediction results over time.

Table 5; Fig. 19 illustrates the CT analysis of the ARDL-FDIAR technique under IEEE39 standard system. In a comparison of CTs for various methods applied to the IEEE39 Standard System, the results are as follows: ARDL-FDIAR attained the fastest time at 11.35 s, followed by CNN at 16.75 s and SGAN + CNN at 16.57 s. Other methods comprises DNN with 22.50 s, SVMGAB at 22.91 s, LSTM at 24.47 s, C-DBN at 24.71 s, and MSA at 28.00 s. The SVM method took 28.48 s, while XGBoost required 32.31 s, and SMAN took 31.91 s. MRFO-DLFDIAD had a CT of 33.42 s, with SGAN + GAL taking 28.90 s and SGAN + GCN the longest at 37.25 s.

Conclusion

In this study, a novel ARDL-FDIAR methodology is presented. The primary goal of the ARDL-FDIAR methodology is to boost security via the FDIA detection process. To accomplish that, the ARDL-FDIAR technique has four stages: z-score normalization, MLOA-based selection feature, IDBN-based FDIA detection, and COA-based parameter selection. At the primary stage, the ARDL-FDIAR approach undergoes Z-score normalization and is deployed to scale the input data. Followed by the attribute reduction process gets invoked using MLOA to choose optimal feature sets. Moreover, the FDIA detection process is performed by modelling the IDBN model. Furthermore, the performance of the IDBN model is improved by the design of a COA-based hyperparameter tuning process. A series of experiments were conducted to ensure the enhancement of the ARDL-FDIAR technique. The results indicated the enhanced security performance of the ARDL-FDIAR technique compared to other DL approaches. The limitations of the ARDL-FDIAR model encompass its reliance on a limited dataset, which may affect the generalizability of the results across diverse contexts or applications. Moreover, the model may face difficulty with high-dimensional data where the curse of dimensionality can

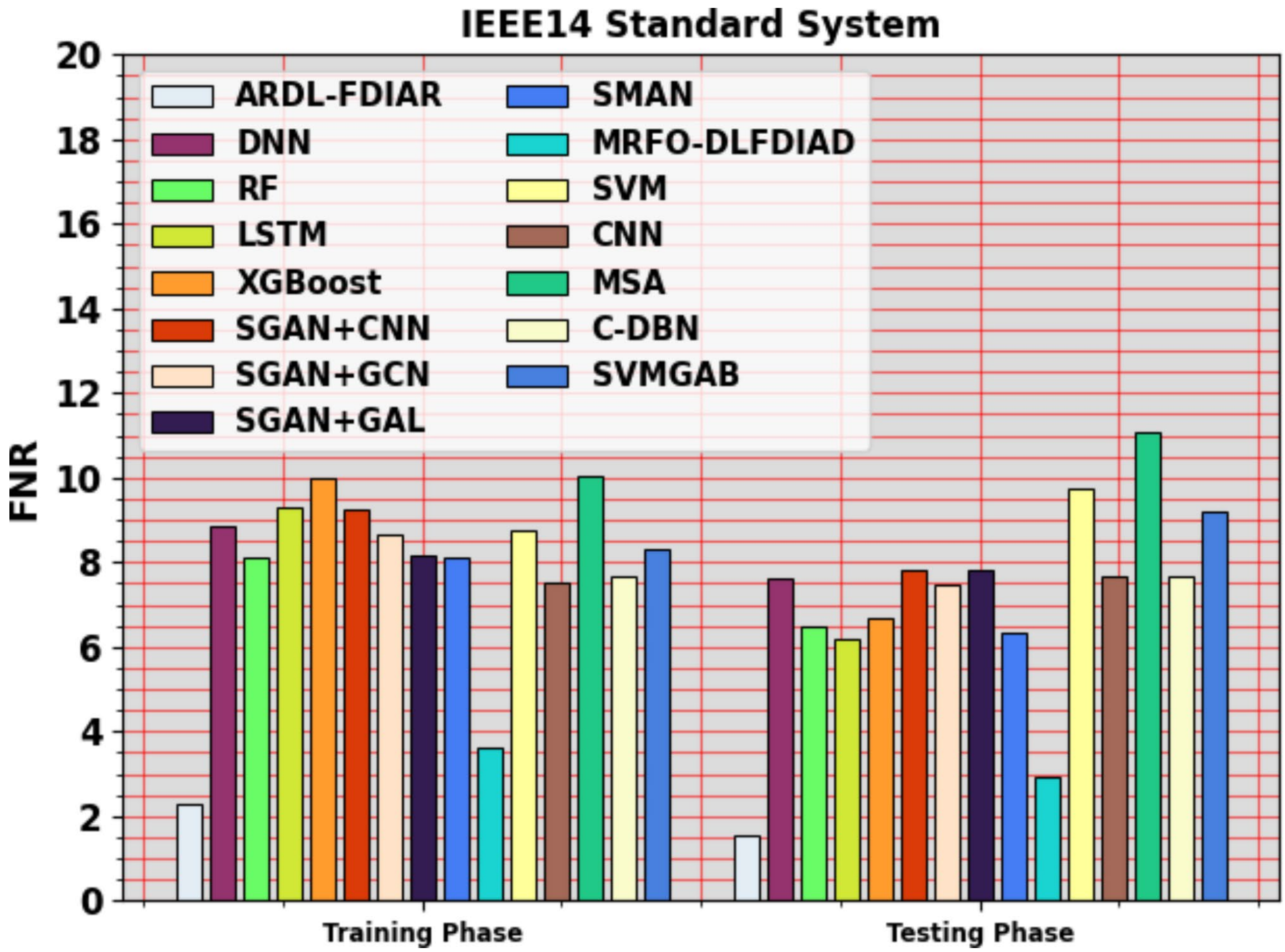

Fig. 8. FNR outcome of ARDL-FDIAR technique under IEEE14 standard system.

impact performance, resulting in overfitting or increased computational complexity. The interpretability of the model's decisions is another concern, as DL methods can often act as "black boxes," making it challenging to understand how specific inputs influence outputs. Finally, the reliance on predefined thresholds in the detection process may result in missed detections or false positives if the thresholds are not optimally set for varying conditions. Future work may concentrate on expanding the dataset to comprise a broader range of scenarios, exploring alternative normalization and feature selection approaches, and investigating the integration of ensemble models to improve robustness. Moreover, real-time implementation and validation of the model in practical environments could give valuable insights for additional enhancements.

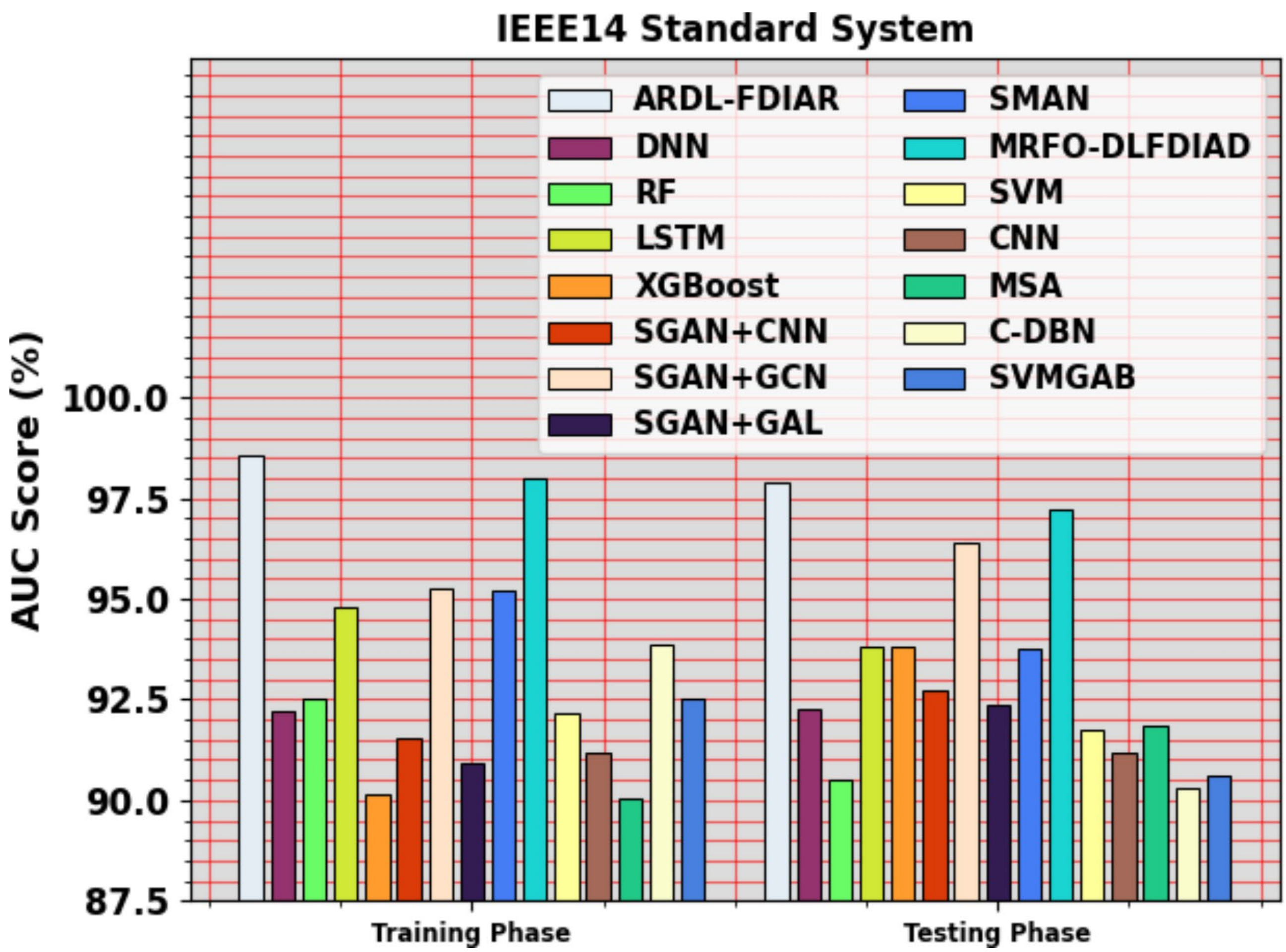

Fig. 9. AUC_{score} outcome of ARDL-FDIAR technique under IEEE14 standard system

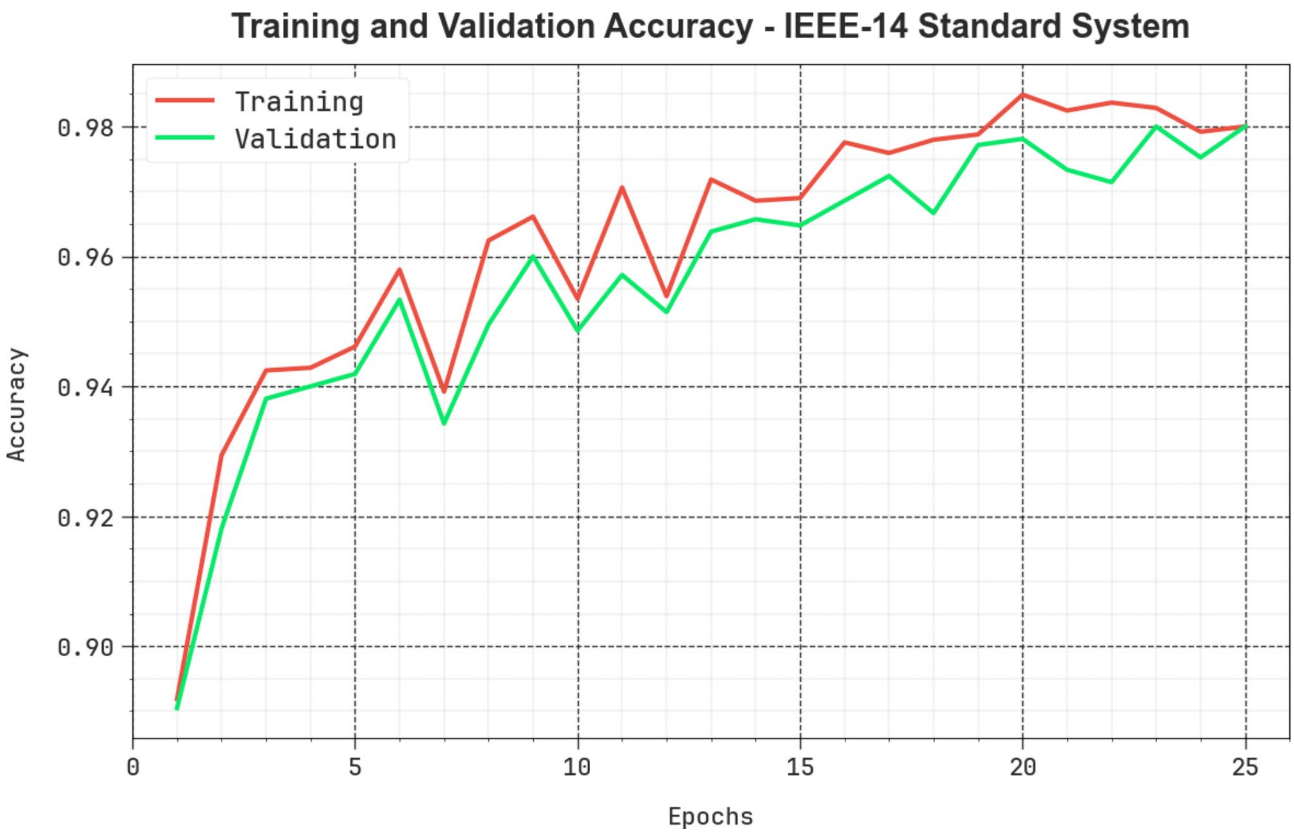

Fig. 10. $Accu_y$ curve of ARDL-FDIAR technique under IEEE14 standard system.

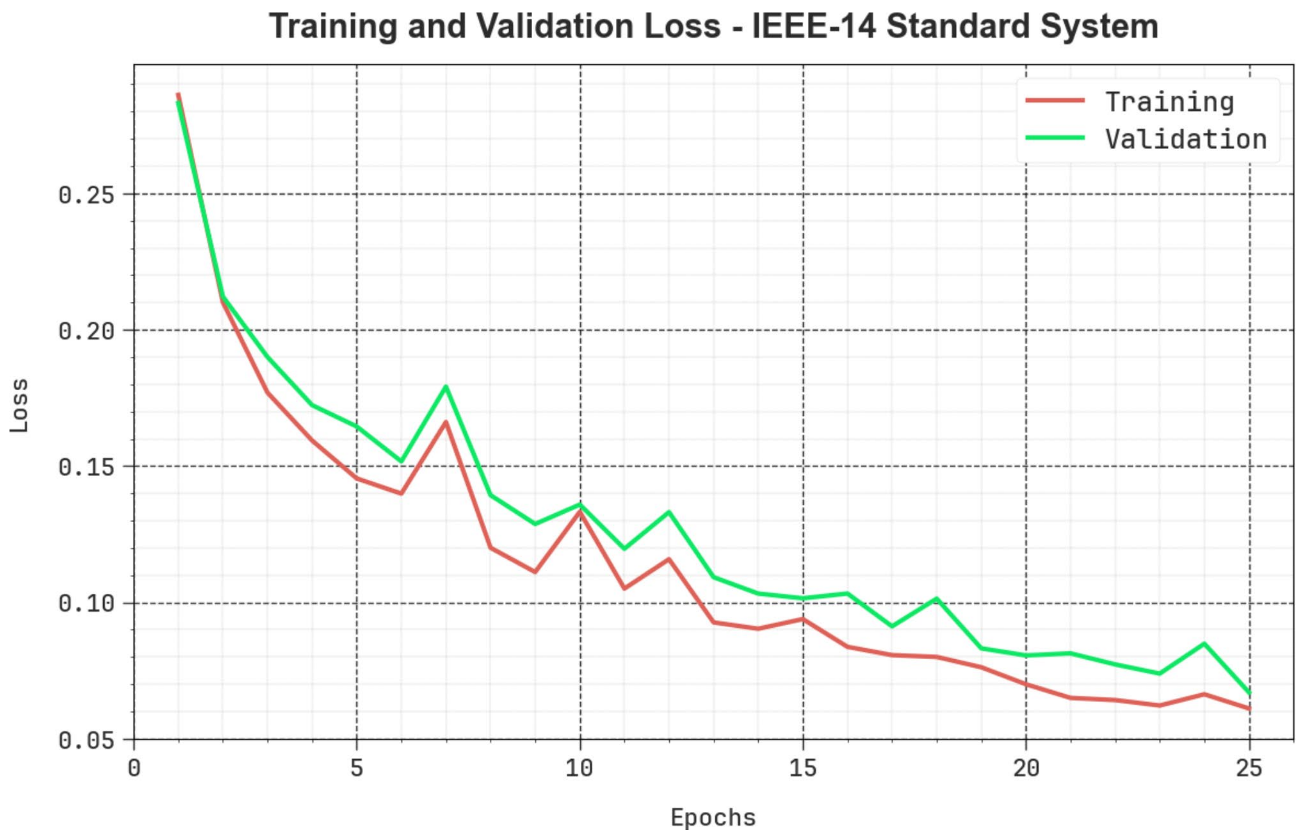

Fig. 11. Loss curve of ARDL-FDIAR technique under IEEE14 standard system.

IEEE14 Standard System	
Methods	**CT (sec)**
ARDL-FDIAR	6.47
DNN	15.92
RF	15.25
LSTM	12.03
XGBoost	9.60
SGAN + CNN	10.95
SGAN + GCN	12.34
SGAN + GAL	10.68
SMAN	9.08
MRFO-DLFDIAD	13.53
SVM	12.67
CNN	18.55
MSA	17.09
C-DBN	11.13
SVMGAB	16.25

Table 3. CT analysis of ARDL-FDIAR technique with other models under IEEE14 standard system.

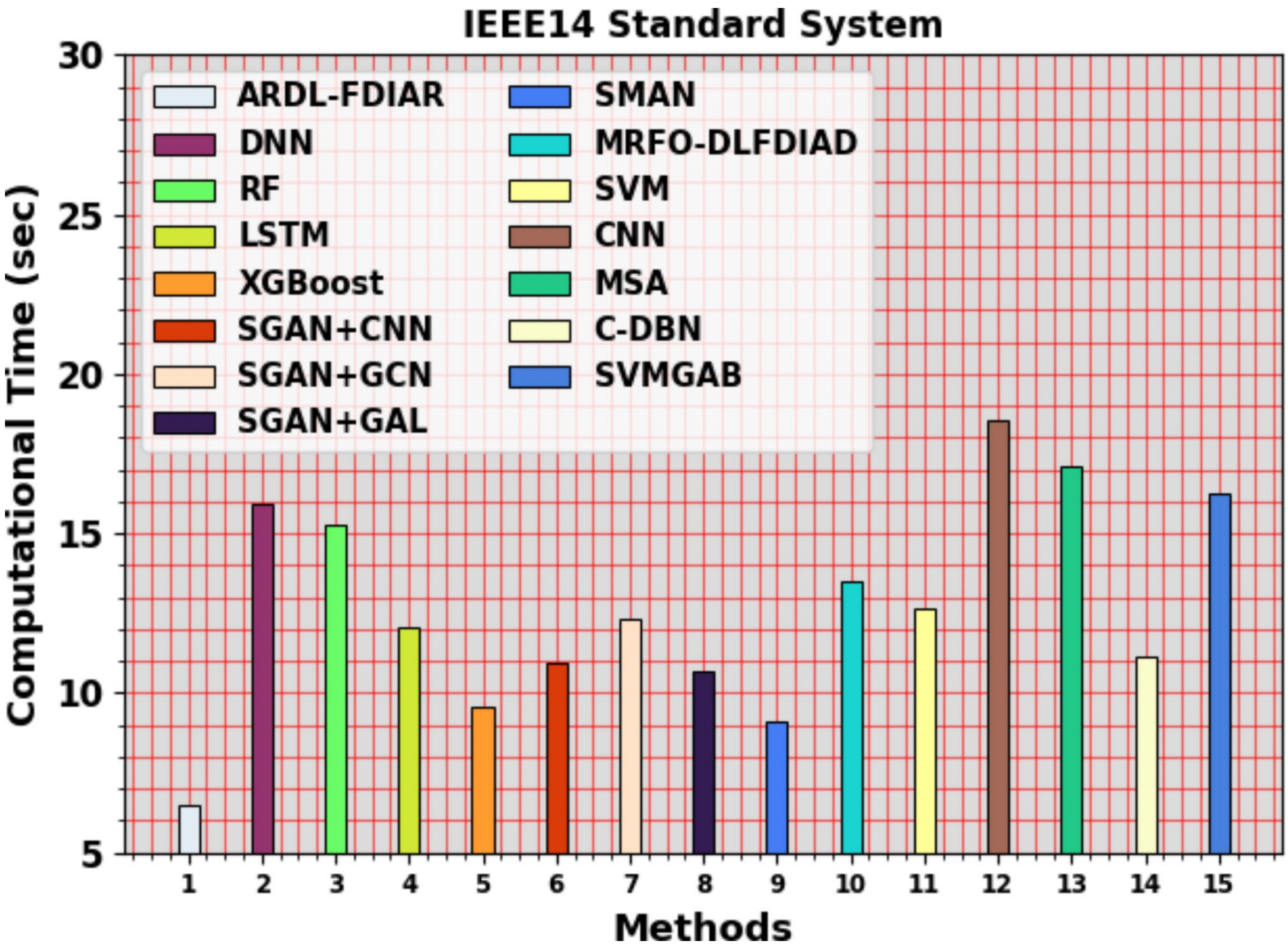

Fig. 12. CT analysis of ARDL-FDIAR technique with other models under IEEE14 standard system.

IEEE39 Standard System				
Methods	$Accu_y$	$Reca_l$	**FNR**	**AUC Score**
Training Phase				
ARDL-FDIAR	98.09	98.27	2.58	99.18
DNN	91.79	95.29	7.23	93.19
RF	93.07	92.27	5.79	95.88
LSTM	93.58	91.21	5.44	92.58
XGBoost	96.03	91.83	5.95	95.23
SGAN + CNN	96.81	95.34	7.23	93.00
SGAN + GCN	92.80	92.69	5.09	97.12
SGAN + GAL	91.63	91.23	5.20	92.73
SMAN	91.94	95.62	5.69	96.89
MRFO-DLFDIAD	97.57	97.67	3.98	98.38
SVM	90.82	92.36	9.19	92.47
CNN	90.10	93.67	7.89	91.58
MSA	90.97	91.08	10.41	90.37
C-DBN	92.38	93.19	7.99	94.12
SVM-GAB	91.35	92.63	8.66	92.92
Testing Phase				
ARDL-FDIAR	97.73	98.00	2.11	98.35
DNN	93.49	94.56	6.38	96.47
RF	94.26	96.59	5.78	91.83
LSTM	96.40	91.80	6.80	94.95
XGBoost	96.63	92.86	5.41	95.01
SGAN + CNN	91.91	96.67	7.00	96.33
SGAN + GCN	92.28	95.75	6.23	95.68
SGAN + GAL	96.96	92.91	5.81	95.48
SMAN	92.57	94.90	5.35	94.46
MRFO-DLFDIAD	97.01	97.23	3.36	97.57
SVM	89.67	93.37	7.41	92.88
CNN	93.82	90.10	10.69	92.97
MSA	91.58	94.00	6.72	90.43
C-DBN	90.89	89.79	10.90	92.08
SVM-GAB	89.91	92.20	8.59	90.36

Table 4. Comparative outcome of ARDL-FDIAR technique with other models under IEEE39 standard system[16,55,56].

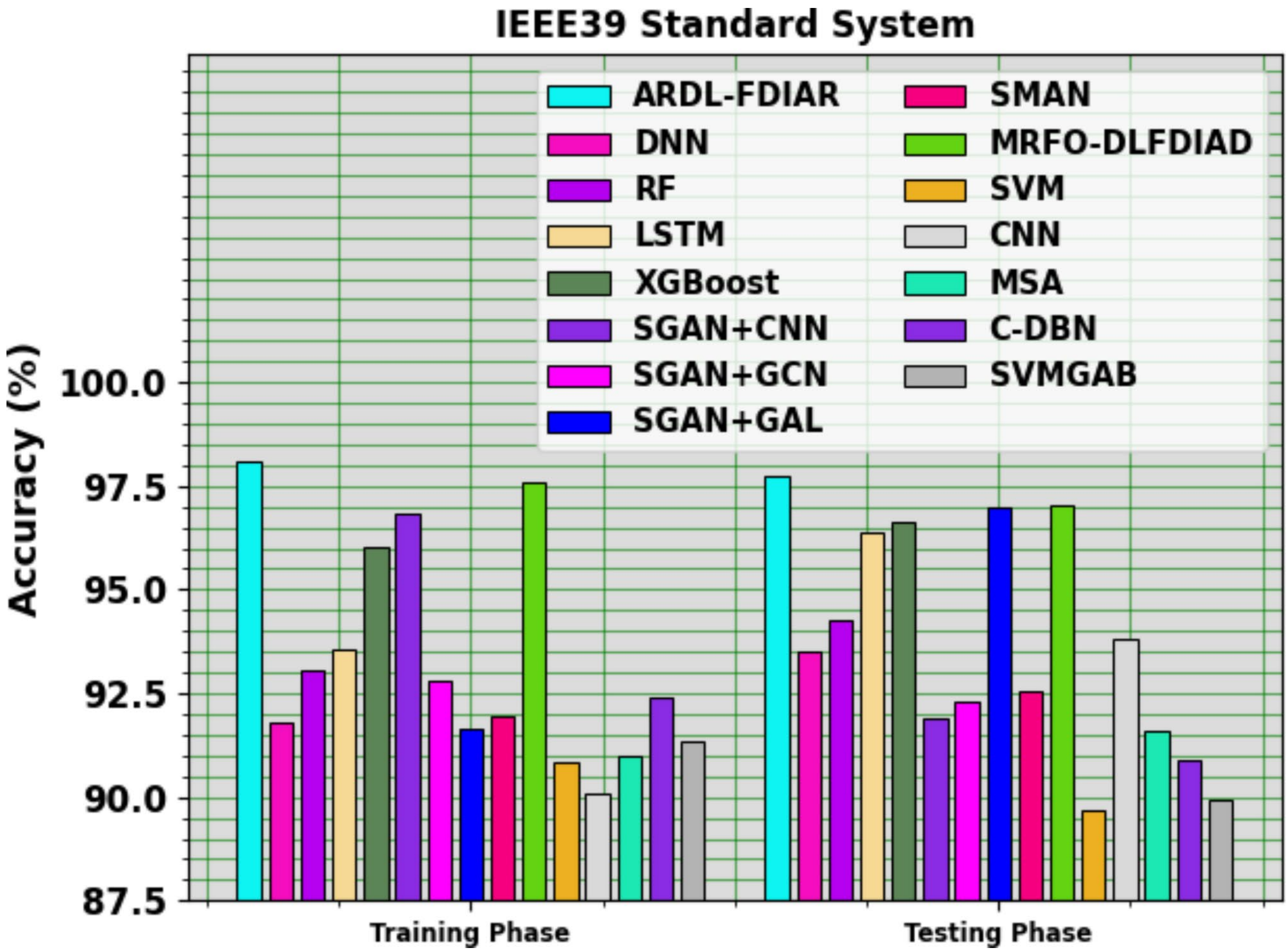

Fig. 13. $Accu_y$ outcome of ARDL-FDIAR technique under IEEE39 standard system

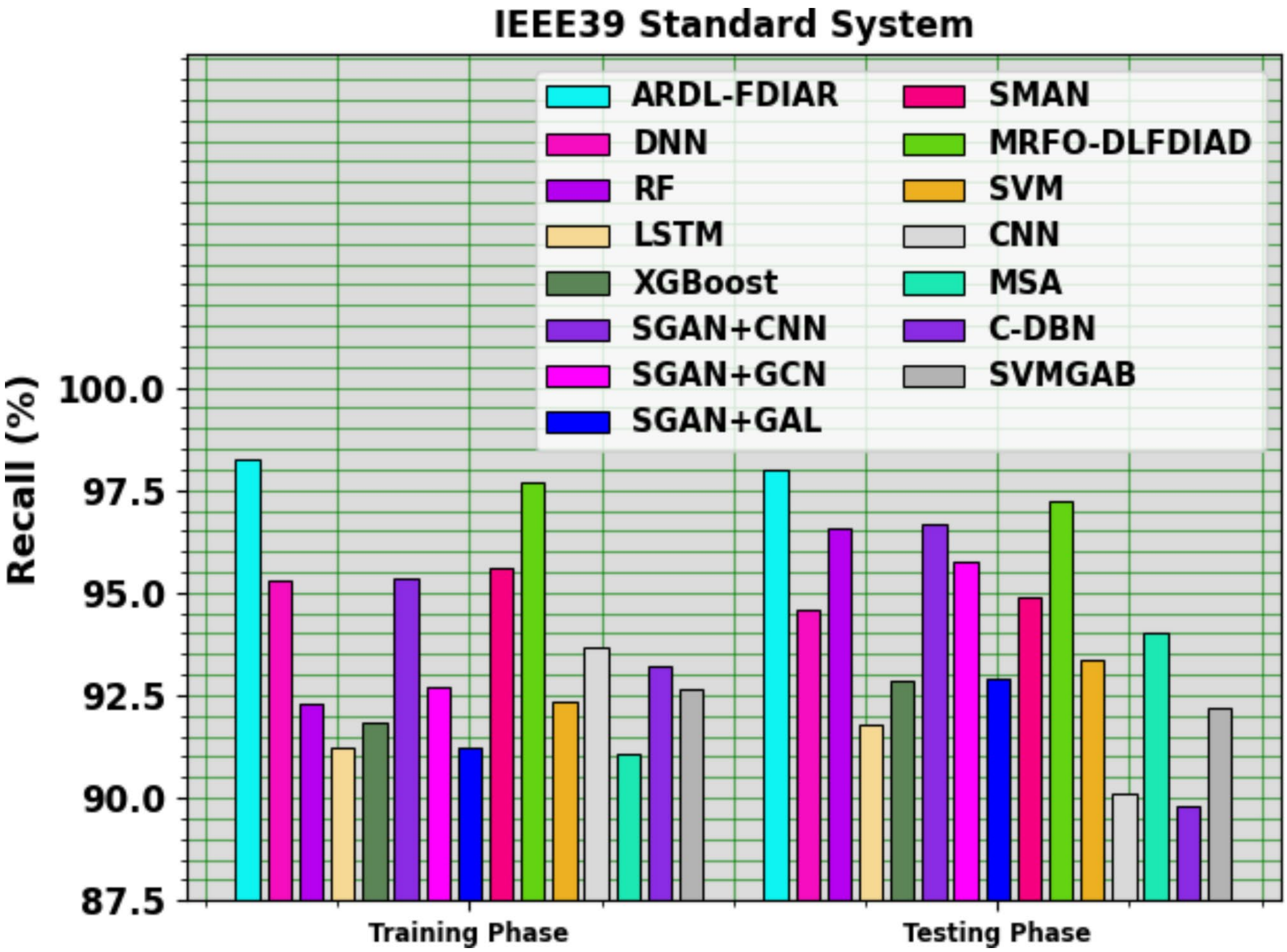

Fig. 14. $Recal$ outcome of ARDL-FDIAR technique under IEEE39 standard system.

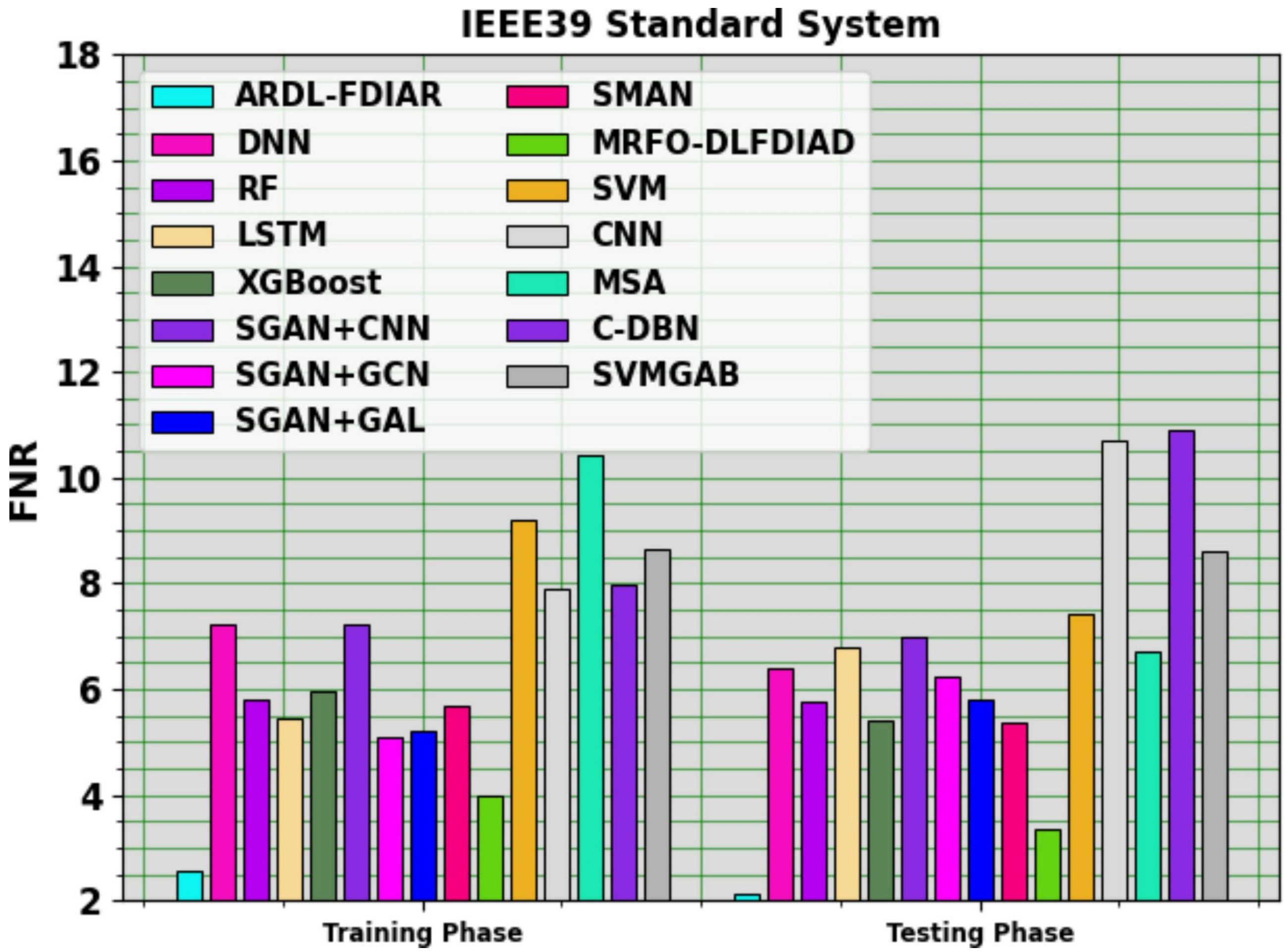

Fig. 15. FNR outcome of ARDL-FDIAR technique under IEEE39 standard system.

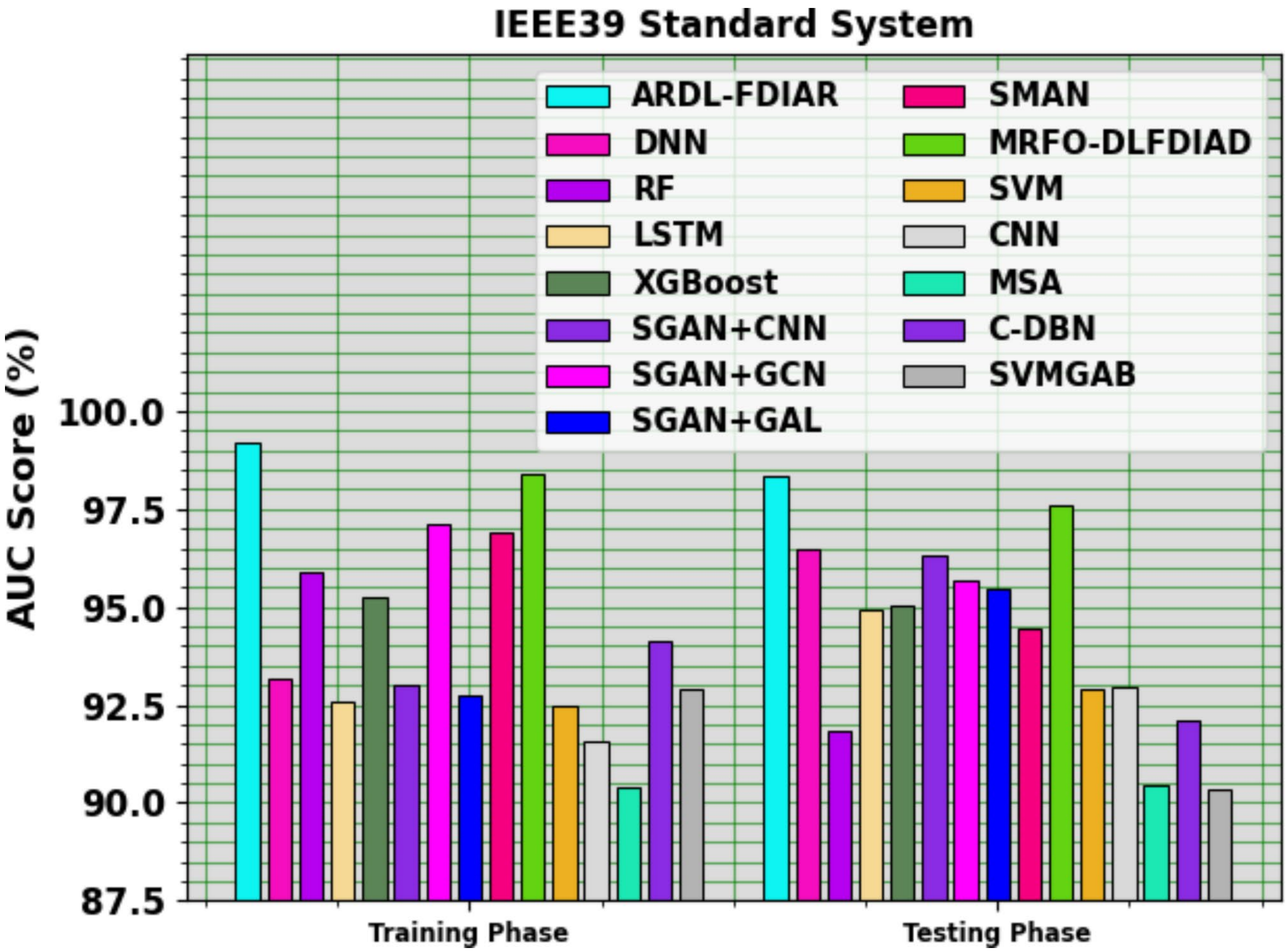

Fig. 16. AUC_{score} outcome of ARDL-FDIAR technique under IEEE39 standard system.

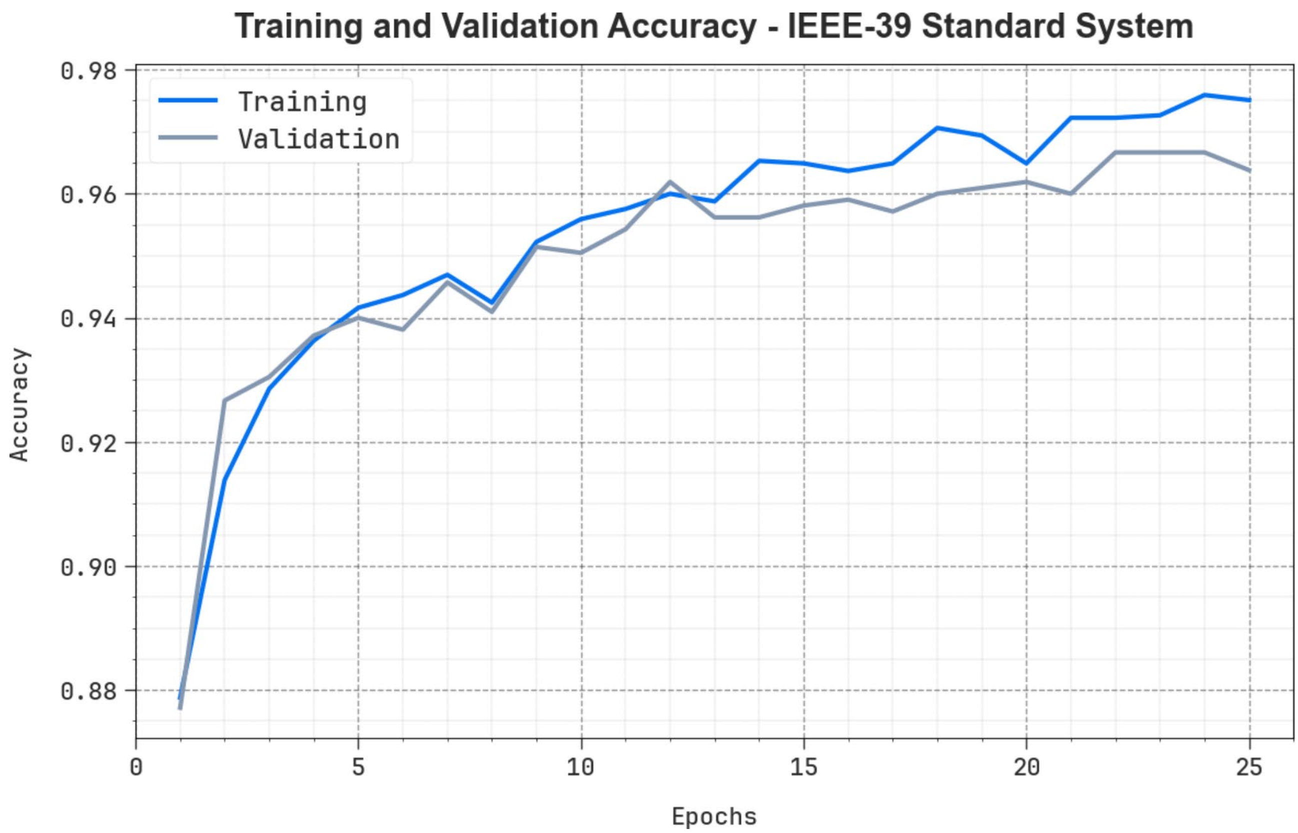

Fig. 17. $Accu_y$ curve of ARDL-FDIAR technique under IEEE39 standard system.

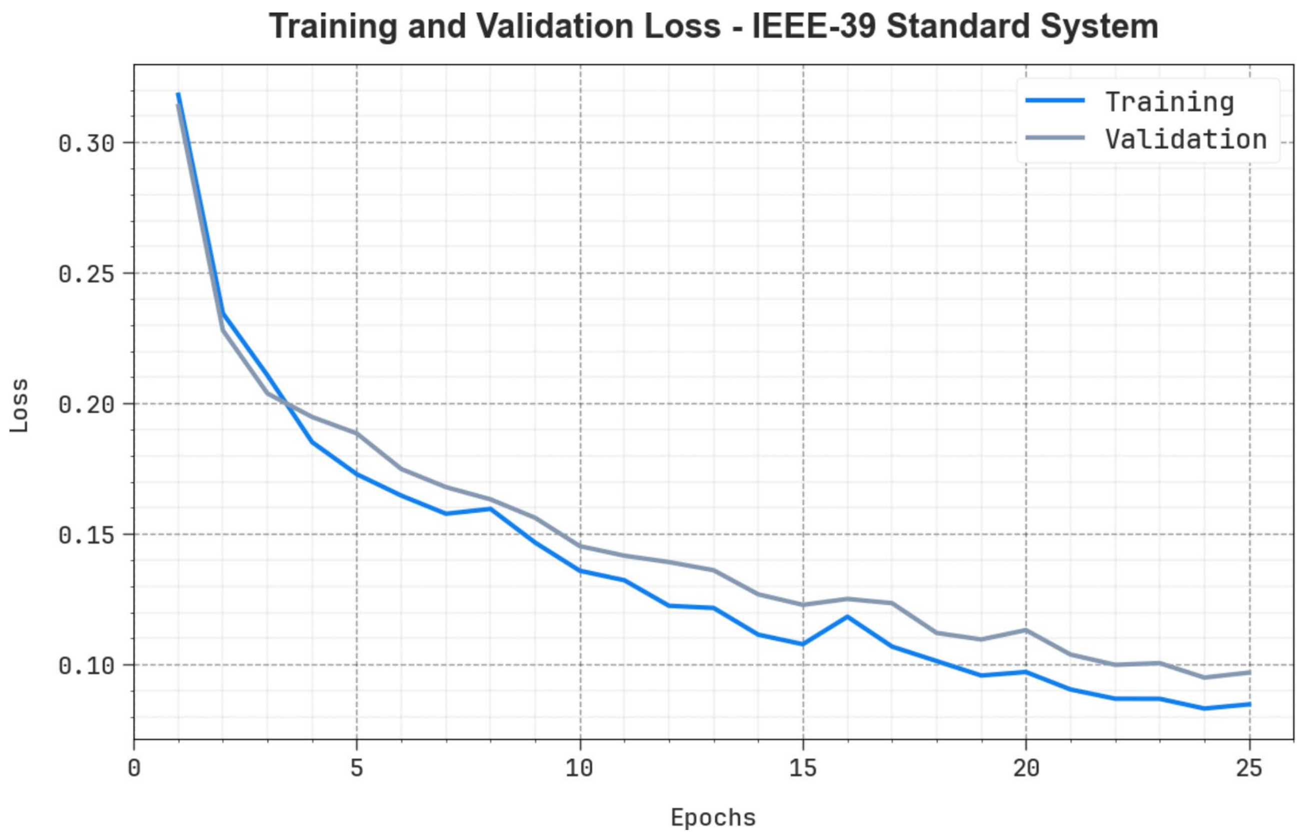

Fig. 18. Loss curve of ARDL-FDIAR technique under IEEE39 standard system.

IEEE39 Standard System	
Methods	**CT (sec)**
ARDL-FDIAR	11.35
DNN	22.50
RF	27.92
LSTM	24.47
XGBoost	32.31
SGAN + CNN	16.57
SGAN + GCN	37.25
SGAN + GAL	28.90
SMAN	31.91
MRFO-DLFDIAD	33.42
SVM	28.48
CNN	16.75
MSA	28.00
C-DBN	24.71
SVMGAB	22.91

Table 5. CT analysis of ARDL-FDIAR technique with other models under IEEE39 standard system.

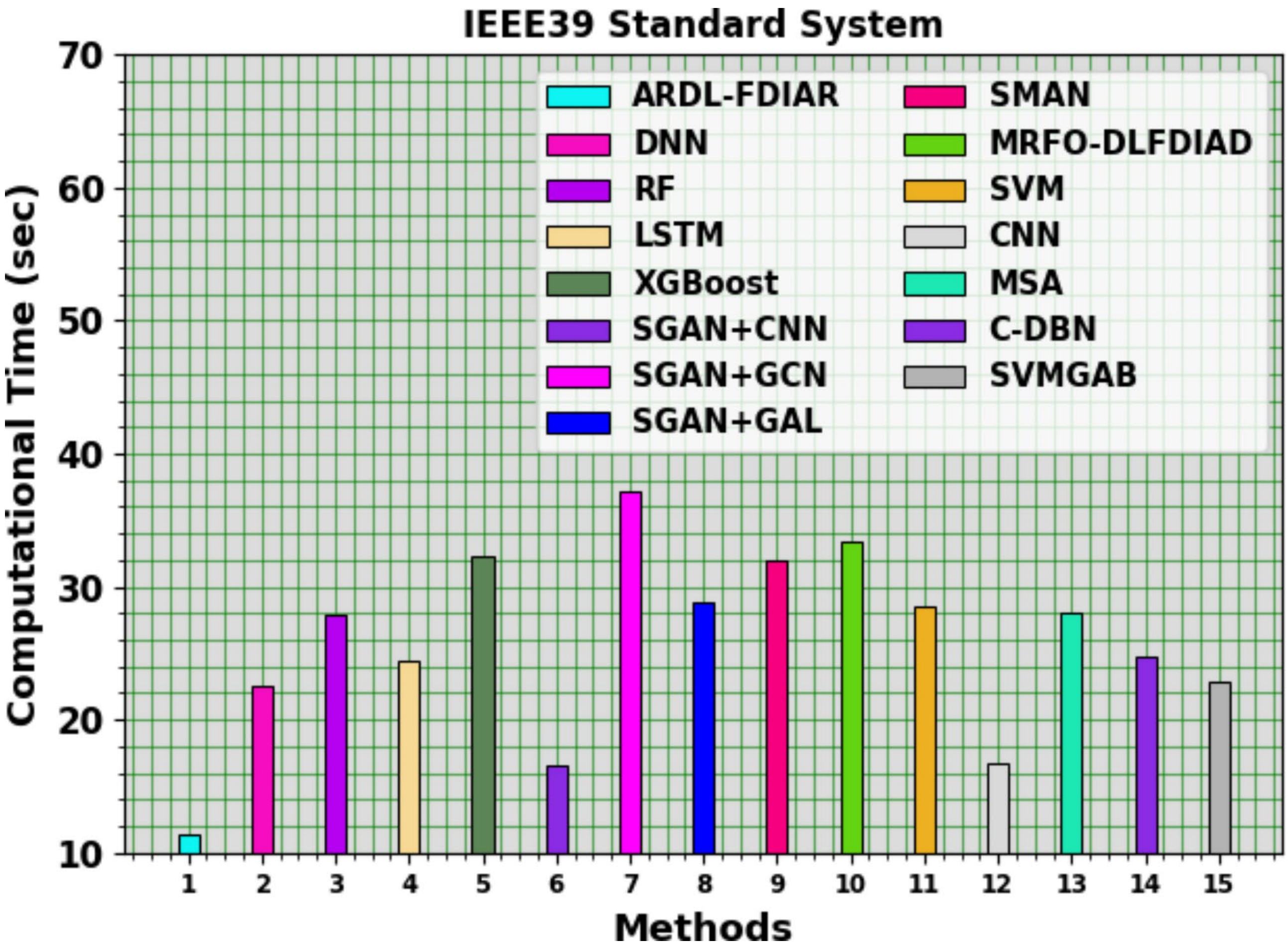

Fig. 19. CT analysis of ARDL-FDIAR technique with other models under IEEE39 standard system.

Data availability
The datasets used and analyzed during the current study available from the corresponding author on reasonable request.

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Author contributions

Conceptualization: Faheed A.F. AlrslaniData curation and Formal analysis: Mohammed Aljebreen, Hamed AlqahtaniInvestigation and Methodology: Faheed A.F. Alrslani, Asma AlshuhailProject administration and Resources: Supervision; 1, Manal Abdullah AlohaliValidation and Visualization: Menwa Alshammeri, Wafa Sulaiman AlmukadiWriting—original draft, Faheed A.F. AlrslaniWriting—review and editing, Manal Abdullah AlohaliAll authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics approval

This article contains no studies with human participants performed by any authors.

Additional information

Correspondence and requests for materials should be addressed to M.A.A.

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